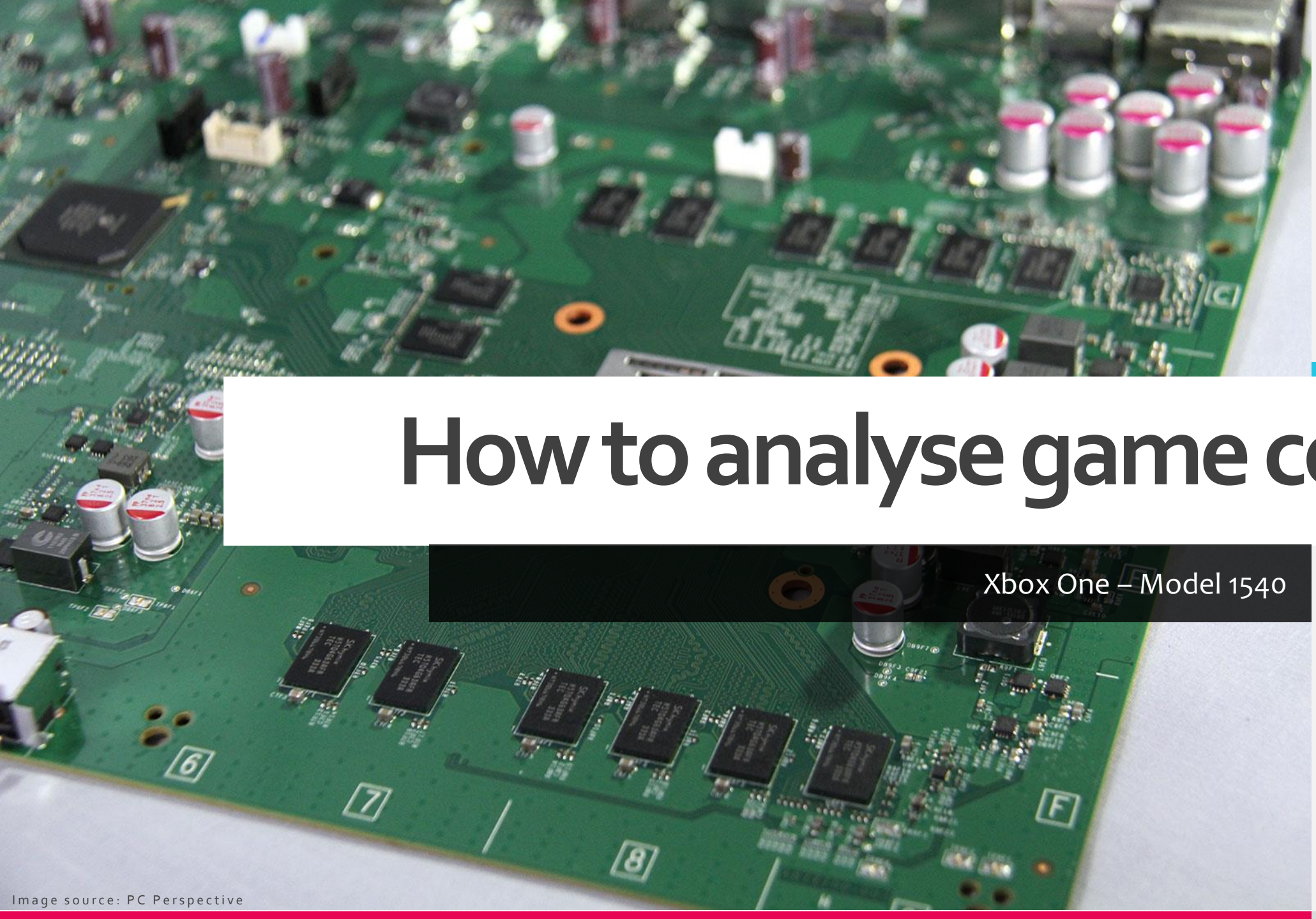


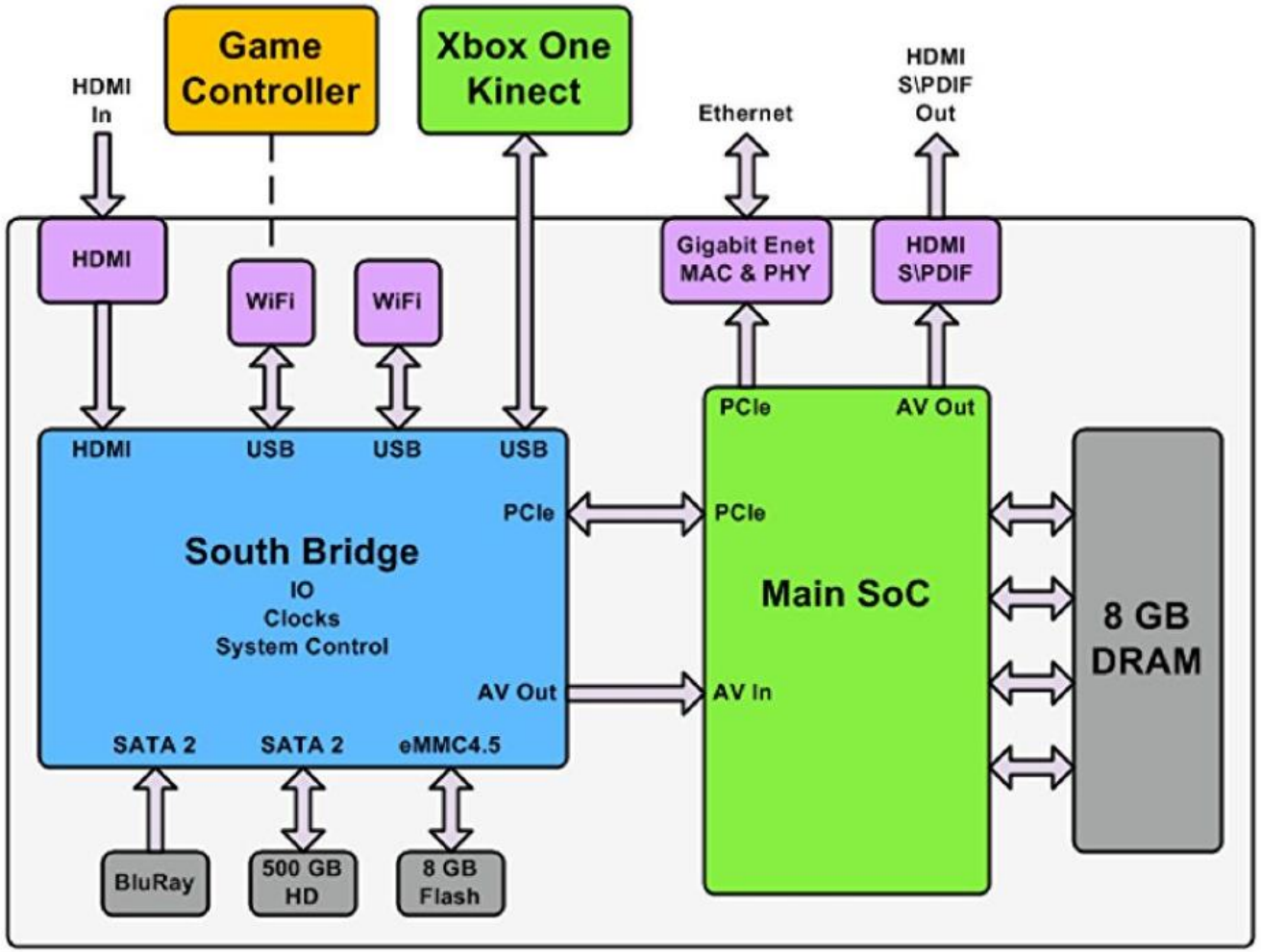
How to analyse game consoles

Xbox One – Model 1540

Image source: PC Perspective

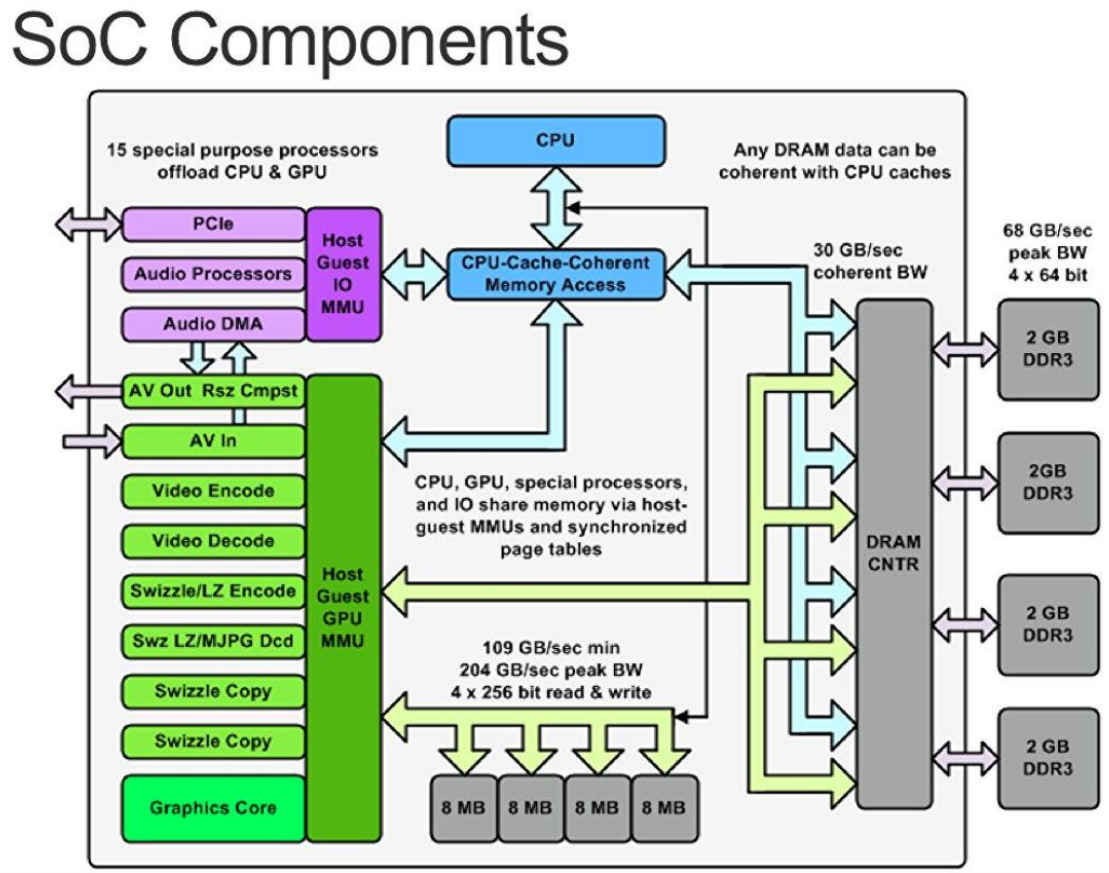
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Xbox One – Main architecture diagram



Xbox One - System overview - SOC

A system-on-chip (SoC) is an integrated circuit that integrates central processing unit (CPU), memory interfaces and the graphics processing unit (GPU) – all on a single substrate or microchip



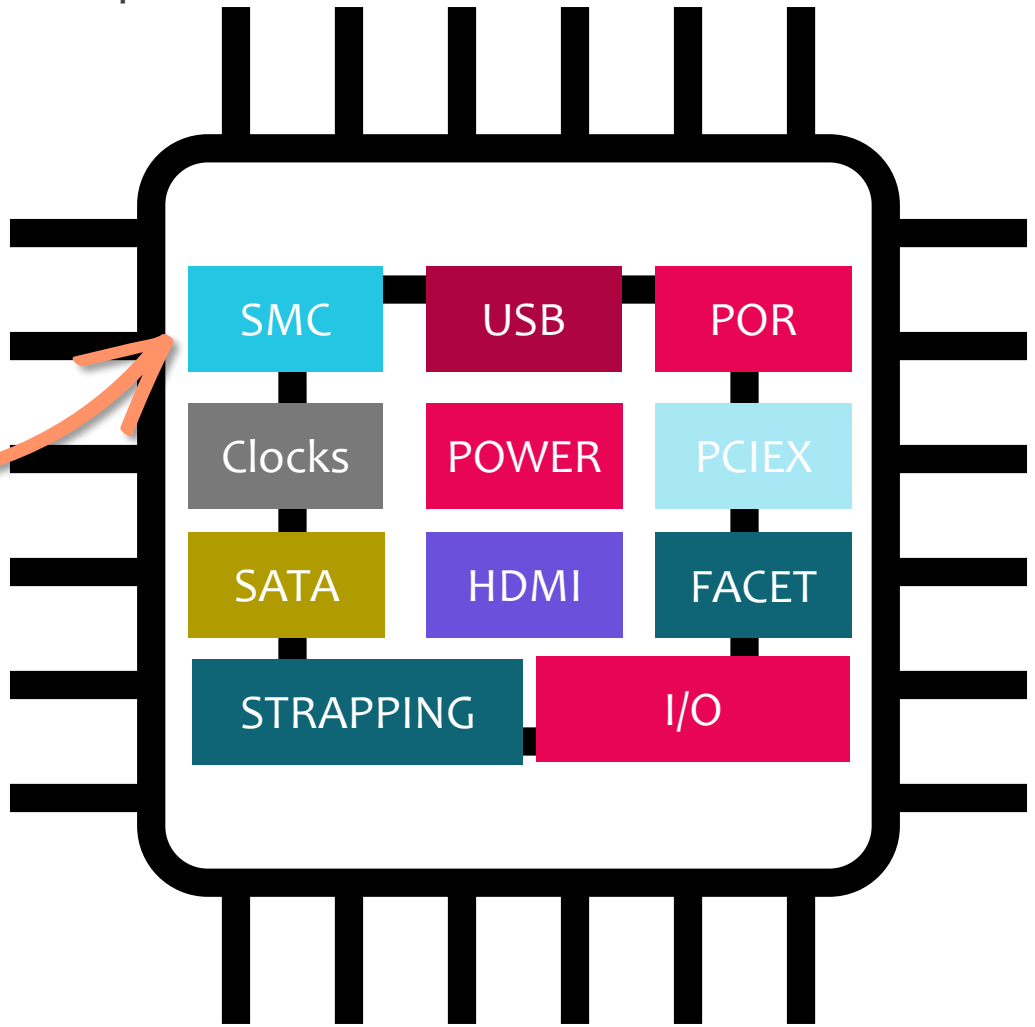


Xbox One - System overview - Southbridge

All I/O operations are handled through a chip called Southbridge The Southbridge connects to all external interfaces. This chip is connected to the SOC chip.

In the Southbridge chip there's a component called System Management Controller (SMC). It draws current even on standby mode.

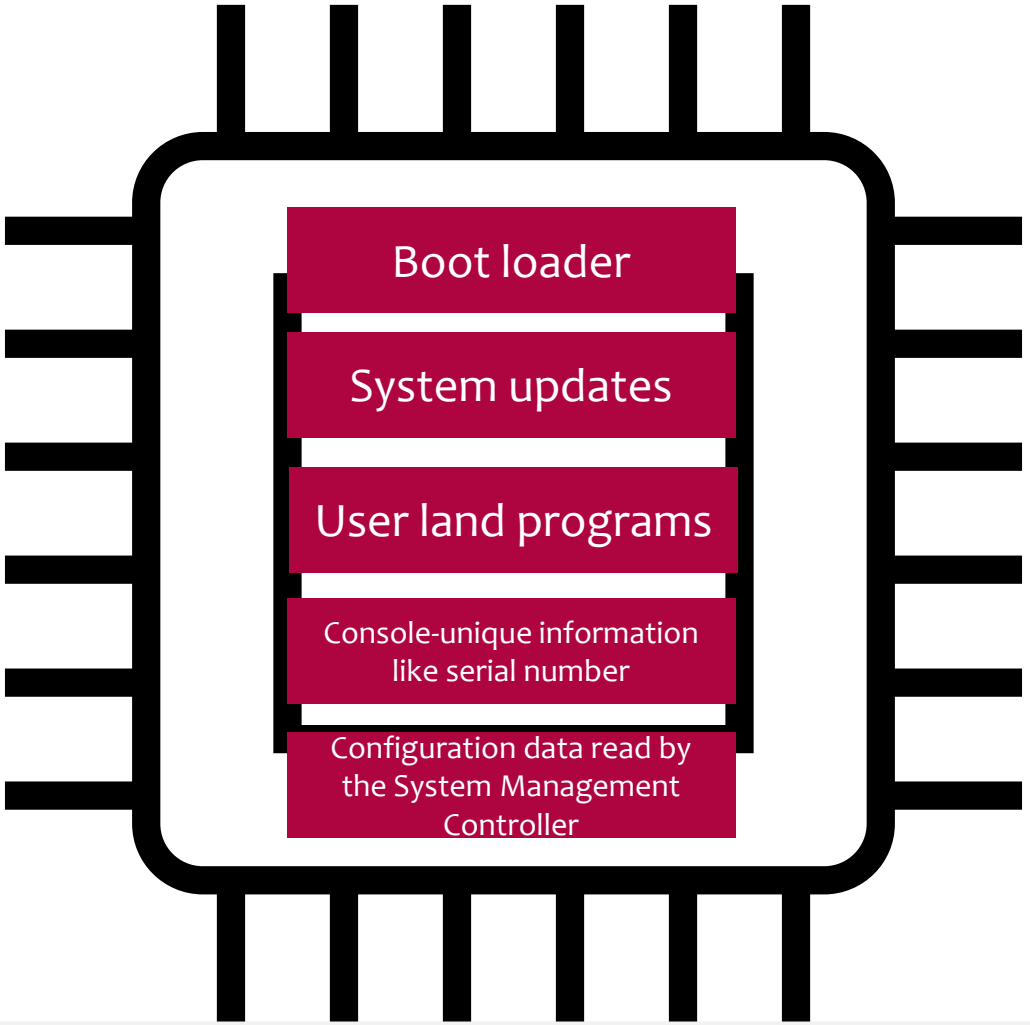
The SMC is responsible for all I/O operations, power management, real-time clock, temperature control, LED control and the infrared sensor





Xbox One - System overview – NAND chip

In the NAND chip the following information is stored





Power Supply

Xbox One Adapter – 220W

Pinout

- GND
- +5V Standby is called: **V_5P0STBY**
- +12V Main is called: **V_12P0**
- Power enable pin named: **PSU_V12P0_EN**



How to analyse standby voltages

Xbox One – Model 1540

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What will happen?

Connecting the power supply to the Xbox One

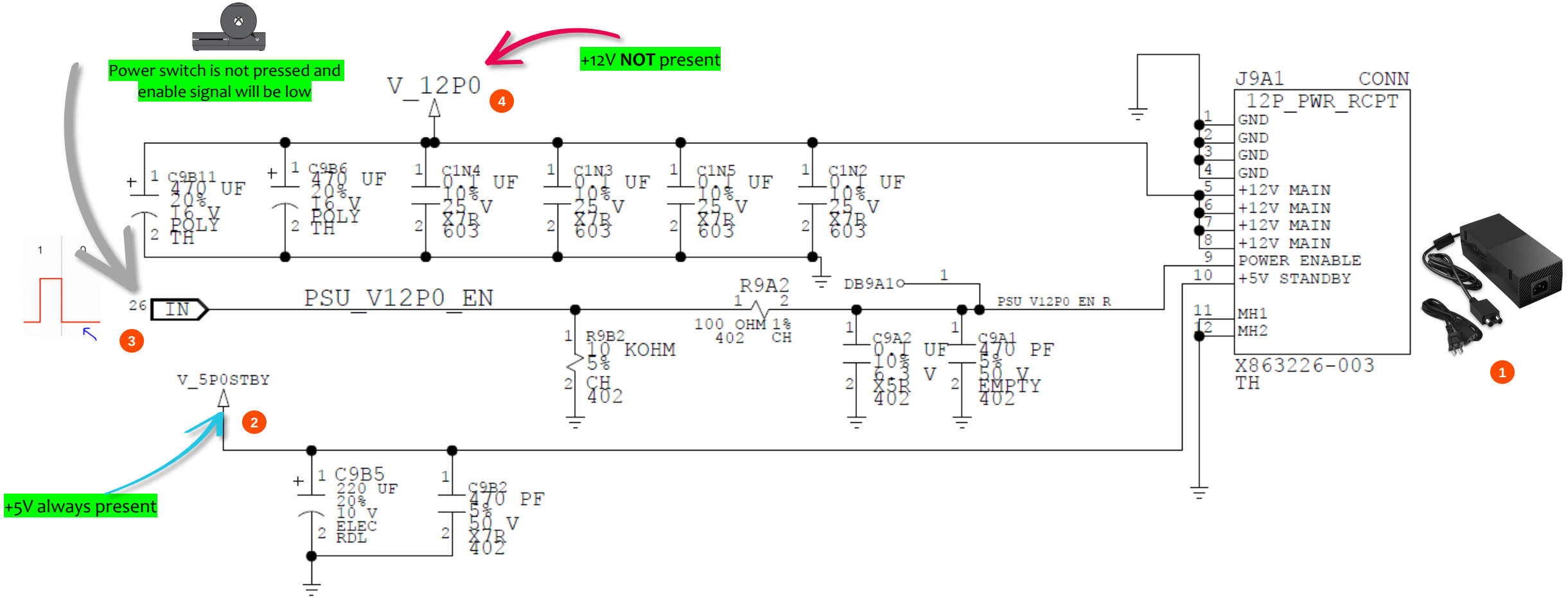


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Power connector – Power switch NOT pressed

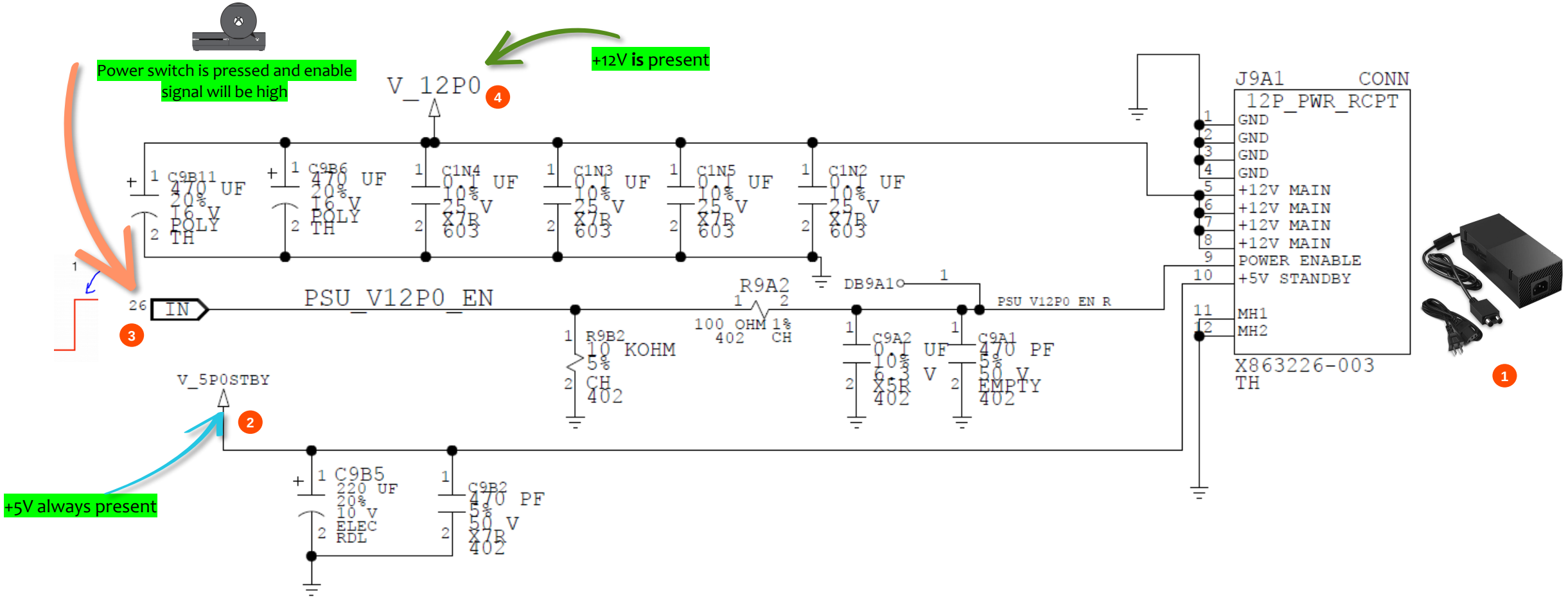
The power supply is connected to the Motherboard and the user didn't press the powerbutton from the frontpanel.

From the motherboard connector the +5V standby voltage will be the main standby powerrail which other standby powerrails will be generated.



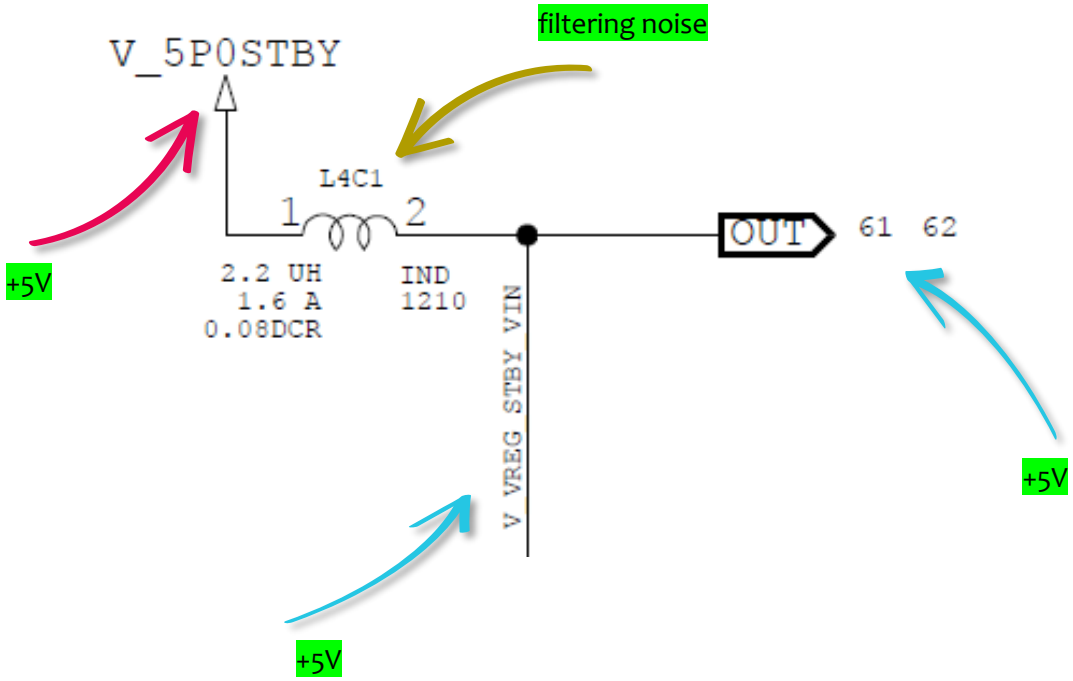
Power connector – Power switch is pressed

When the user press the power switch the system will check if all signals and powerrails are ok and present. Then it will set the enable signal PSU_V12P0_EN to HIGH.



Powerrails with different names

When a branch is created from a primary powerrail the name of the secundary rail can be different and the voltage can be different



From the **V_5P0STBY** voltage rail there is a branch which is called **V_VREG_STBY_VIN** both powerrails are **+5V**

Step 1: +1.1V standby voltage

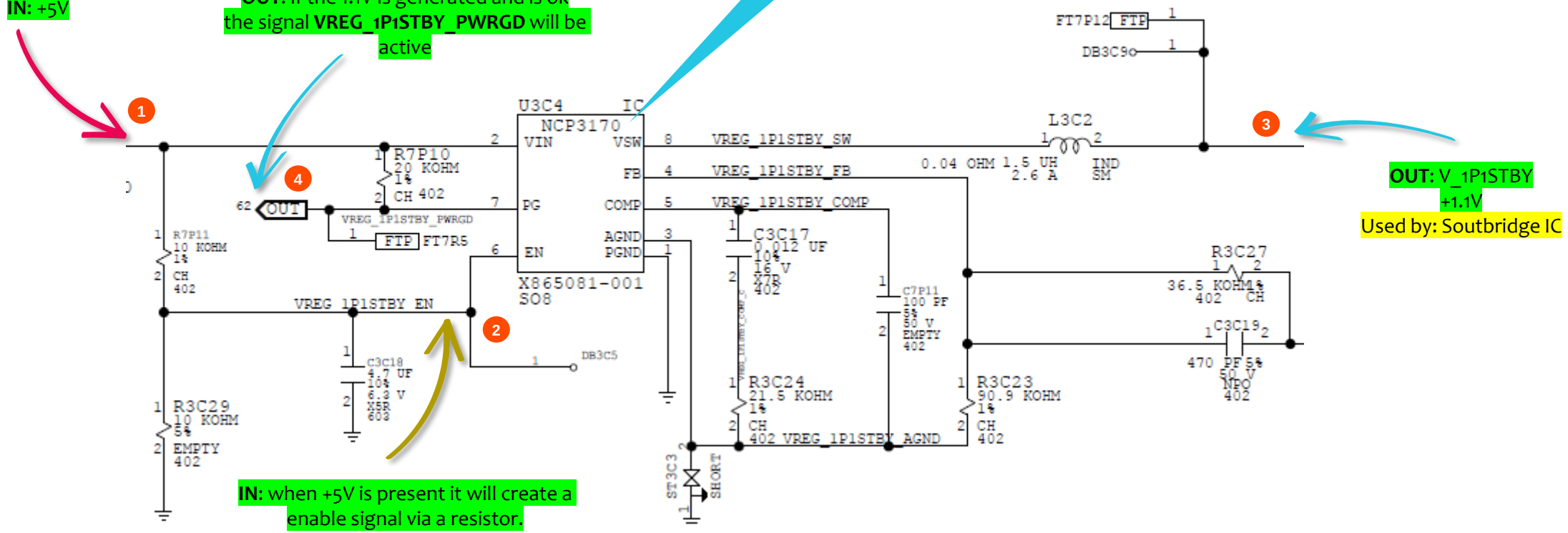
When the +5V standby is present it has to create the 1.1V standby (**V_1P1STBY**) power rail.

The NCP3170 is a PWM Switching Buck Regulator. The NCP3170 operates from 5V and is producing 1.1V output voltage

1 From the **V_VREG_STBY_VIN** (**+5V**) power rail, the 1.1V standby power rail is created called **V_1P1STBY** (**+1.1V**)

IN: +5V

OUT: if the 1.1V is generated and is ok the signal **VREG_1P1STBY_PWRGD** will be active



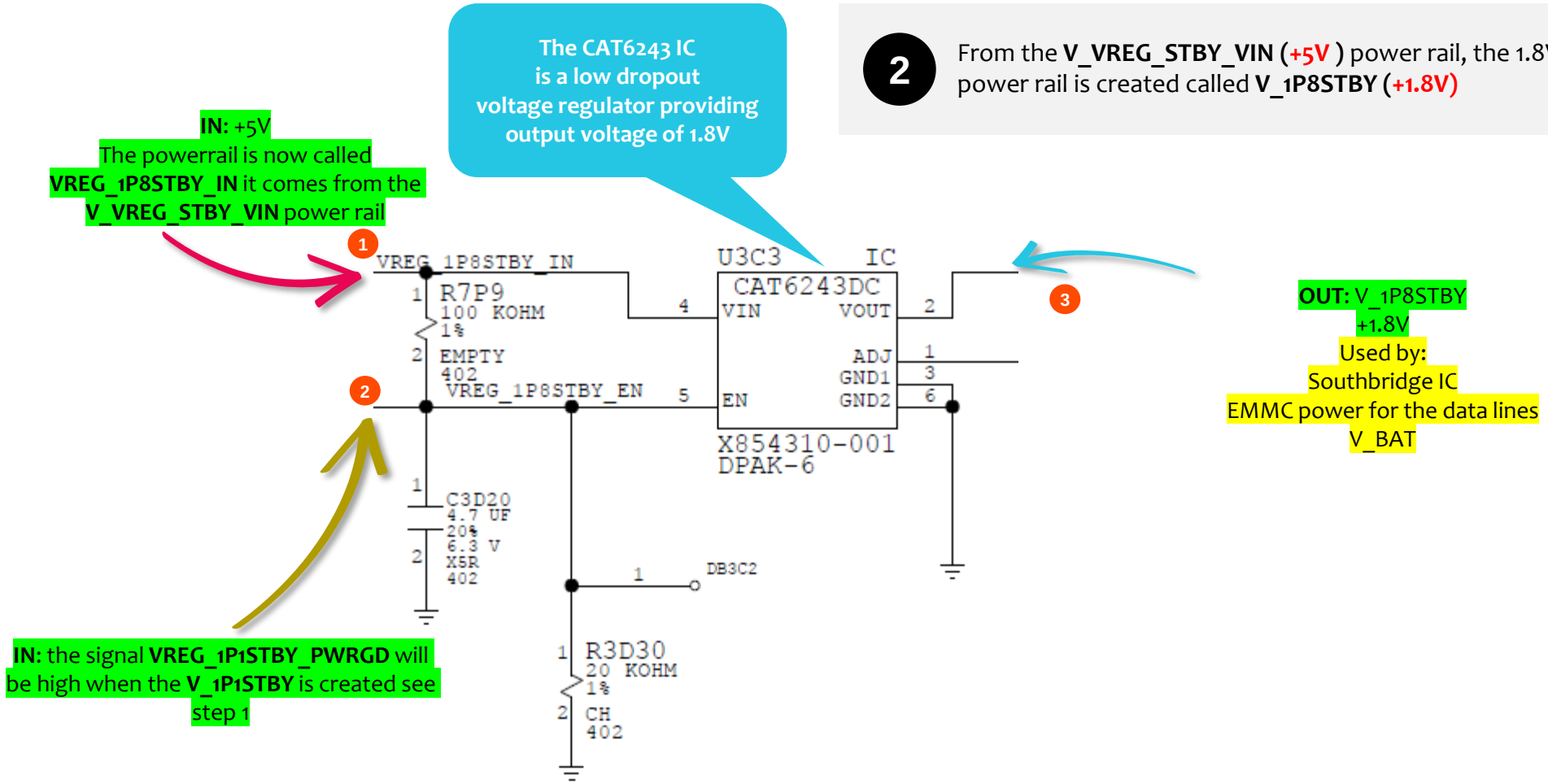
OUT: V_1P1STBY
+1.1V
Used by: Soutbridge IC

IN: when +5V is present it will create a enable signal via a resistor.

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Step 2: +1.8V standby voltage

When the +5V standby is present it has to create the 1.8V standby (**V_1P8STBY**) power rail.

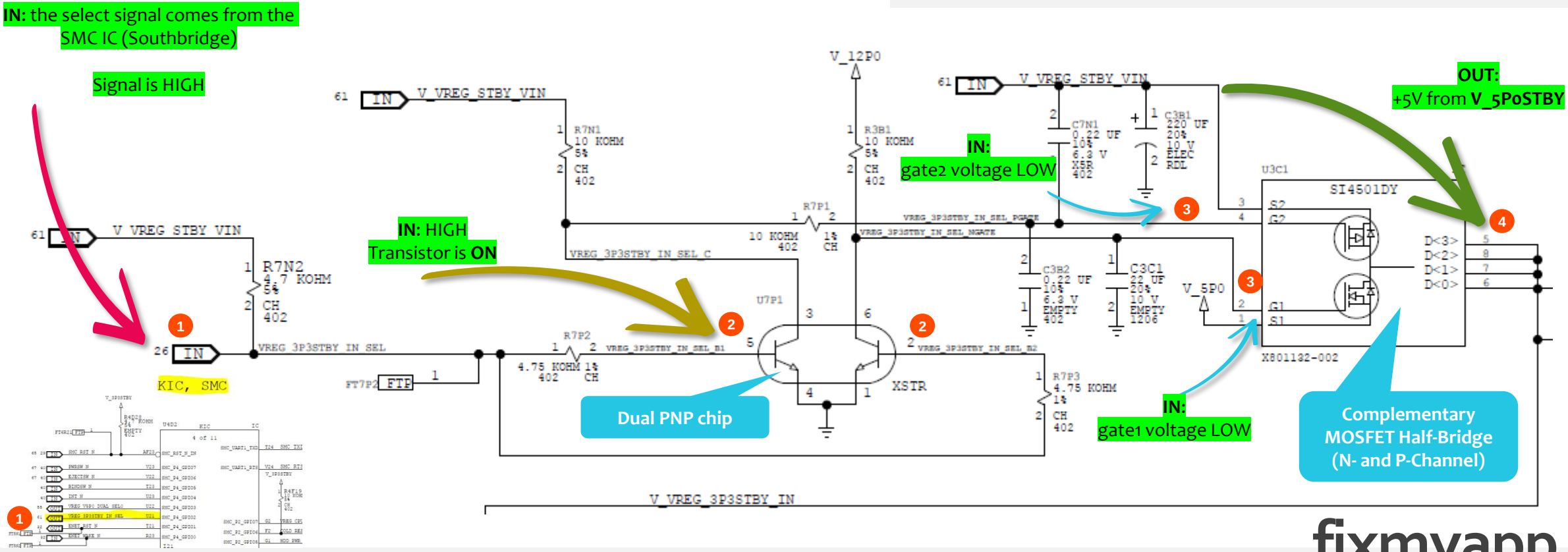


Step 3: Voltage switchers for VREG_3P3STBY_VCC

To power the chip that creates the 3.3V standby (**V_3P3STBY**) voltage the SMC has to send an enable / select signal to the voltage switcher circuit to determine from wich voltage source will deliver +5V.

When enable signal **VREG_3P3STBY_IN_SEL** is HIGH then the +5V source will be **V_5P0STBY (standby)**

3 From the **+5V** voltage rail, the 3.3V standby powerrail is created called **V_3P3STBY (+3.3V)**



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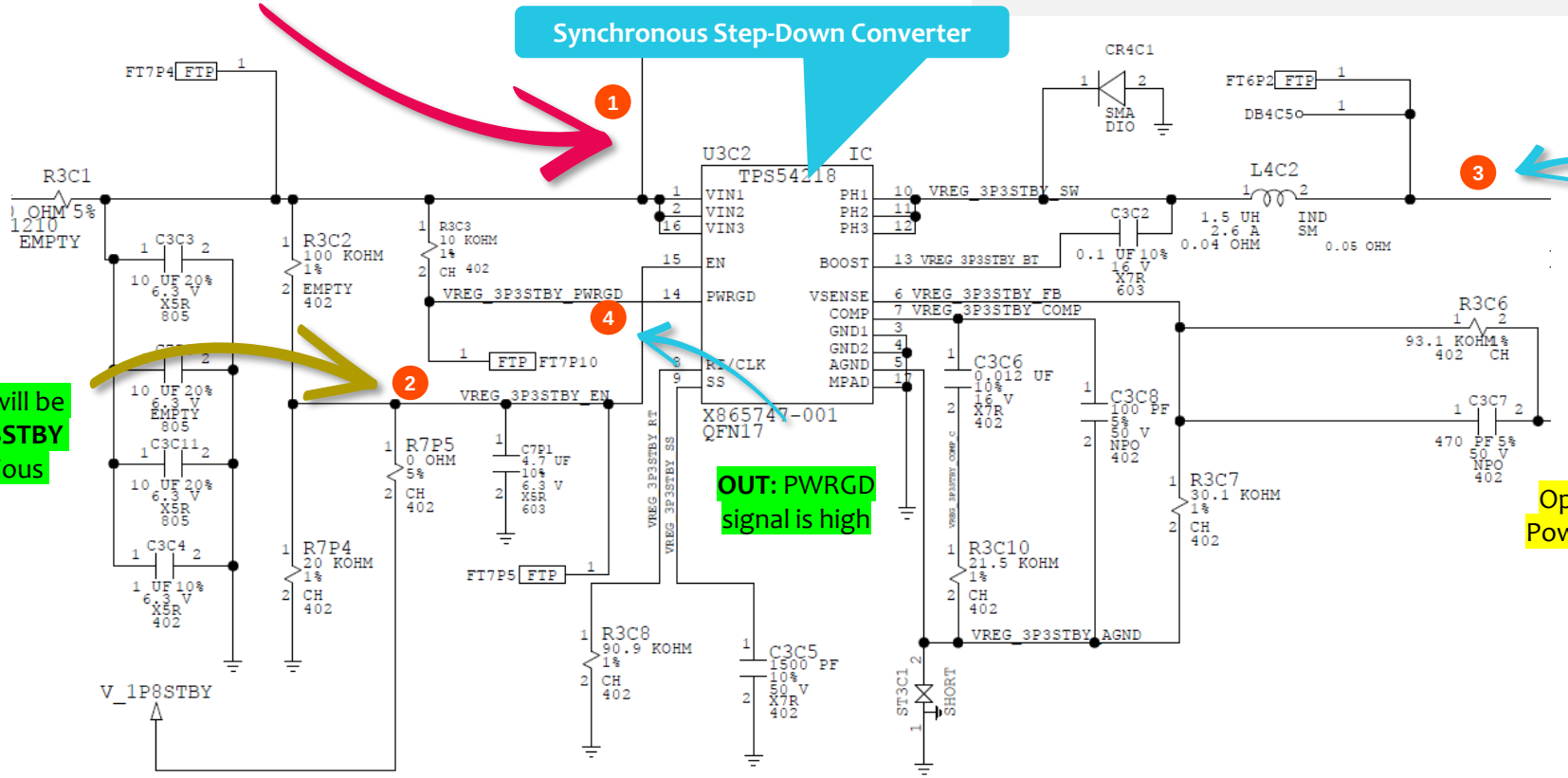
Step 4: +3.3V standby voltage

When the +5V is present it has to create the **V_3P3STBY** voltage rail. This is a +3.3V standby powerrail.

IN: +5V from **V_VREG_3P3STBY_IN**
This voltage can come from two sources: **V_VREG_STBY_VIN** or **V_5P0**
this is determant by some switch circuit in step 3

IN: the signal **VREG_3P3STBY_EN** will be high when the **V_1P8STBY** is created see previous sheet

4 From the **+5V** voltage rail, the 3.3V standby powerrail is created called **V_3P3STBY (+3.3V)**

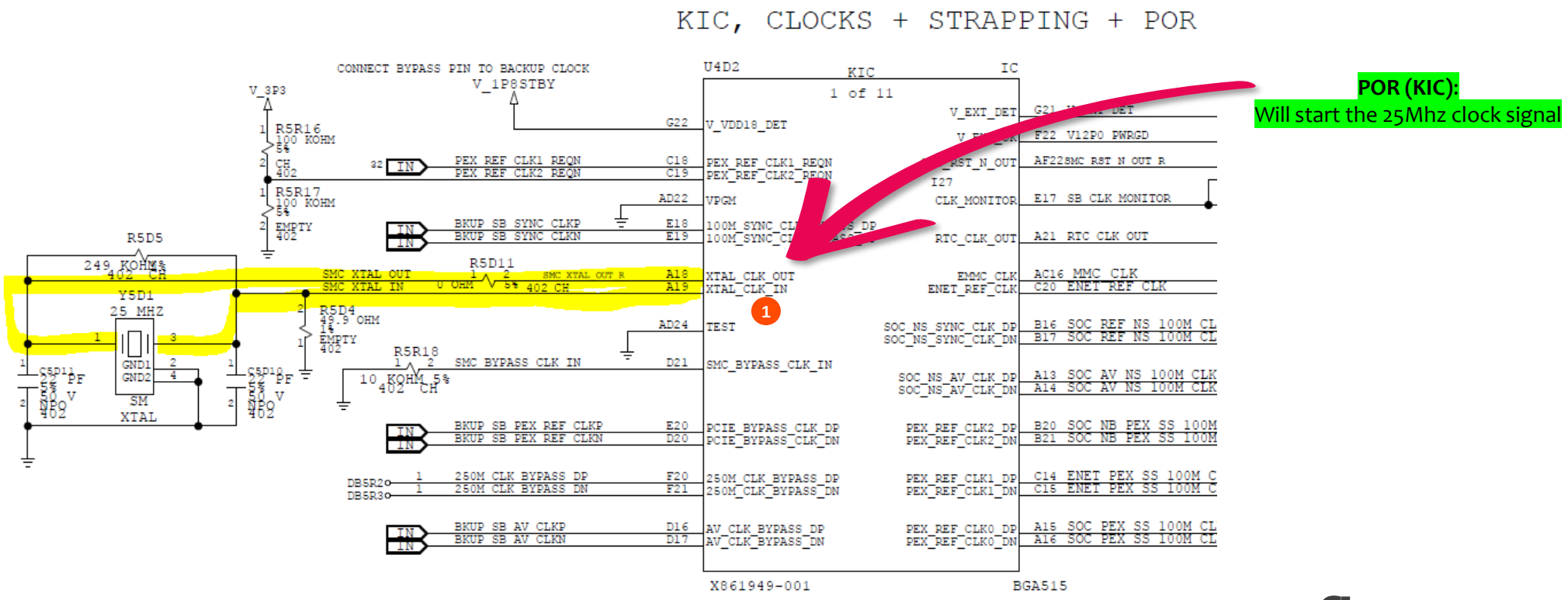


OUT: PWRGD signal is high

OUT: V_3P3STBY
+3.3V
Used by:
Soutbridge XTAL
Southbridge
Ethernet controller
EMMC Memory
WiFi
Optical drive awake signal
Power / Eject / Sync / IR data lines
IR sensor
Eeprom / bios ic

Step 5: Generating 25Mhz clock signal

5 The POR (KIC) will start the clock signal. This clock signal will be send to the SMC (KIC). The clock needs a voltage of +3.3V to operate. **V_KIC3P3XTAL**

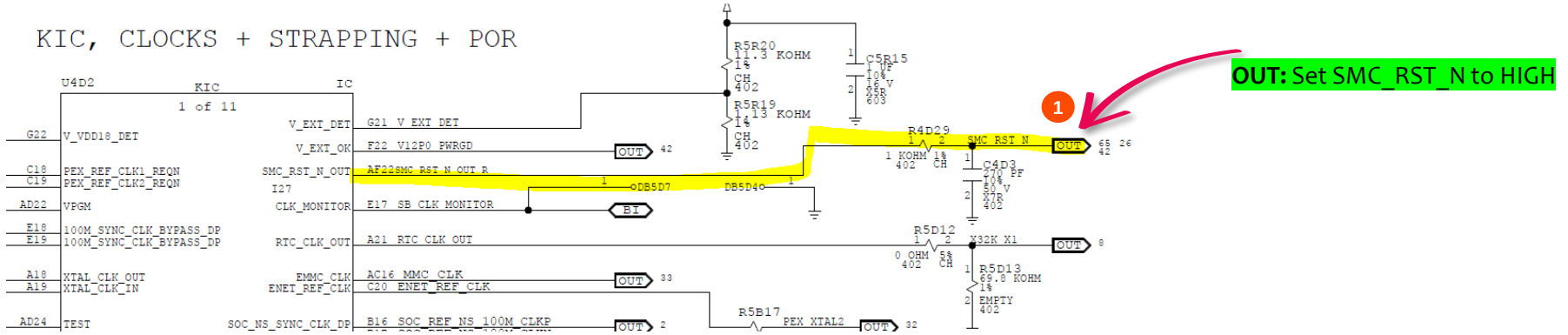


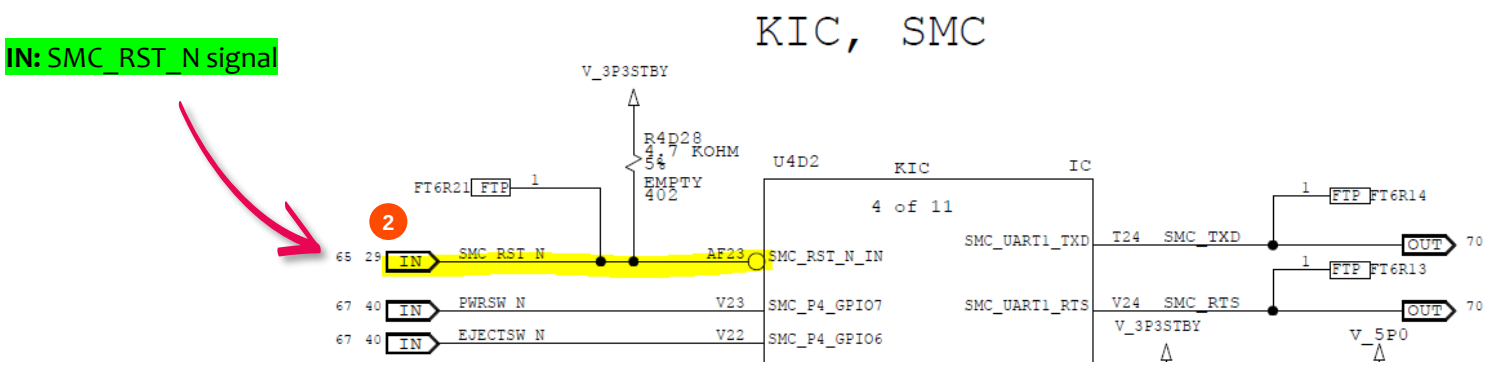
POR (KIC):
Will start the 25Mhz clock signal

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Step 6: Send SMC Reset signal

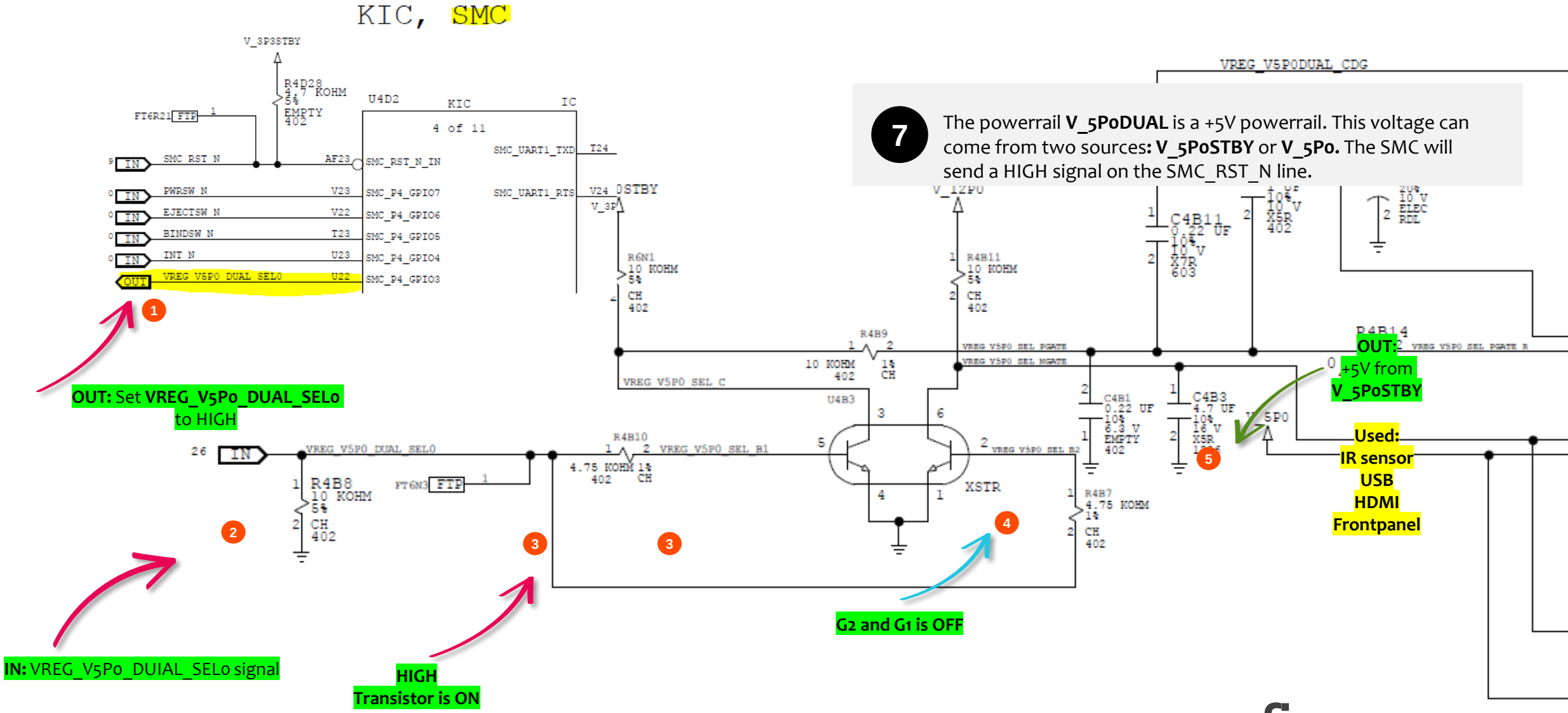
6 The POR (KIC) will set the signal SMC_RST_N to HIGH and sends it to the SMC (KIC)





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Step 7: Voltage switchers for V_5PoDUAL



7 The powerrail **V_5PoDUAL** is a +5V powerrail. This voltage can come from two sources**: V_5PoSTBY** or **V_5Po.** The SMC will send a HIGH signal on the SMC_RST_N line.

1 **OUT:** Set **VREG_V5Po_DUAL_SELo** to HIGH

2 **IN:** VREG_V5Po_DUIAL_SELo signal

3 **HIGH**
Transistor is ON

4 **G2 and G1 is OFF**

5 **OUT:** +5V from **V_5PoSTBY**

Used:
IR sensor
USB
HDMI
Frontpanel

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Standby

Xbox is now in standby mode and waiting for a instruction



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How to analyse power on voltages

Xbox One – Model 1540

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Power On

What will happen when you press the power switch?

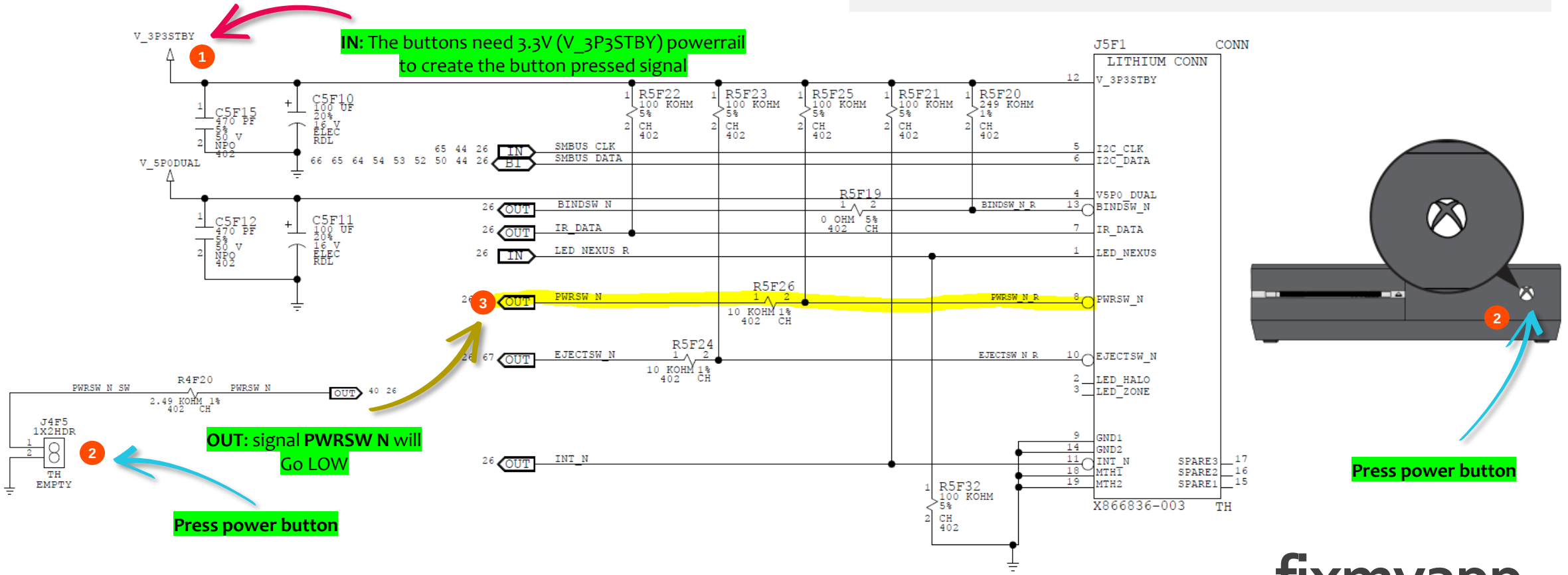


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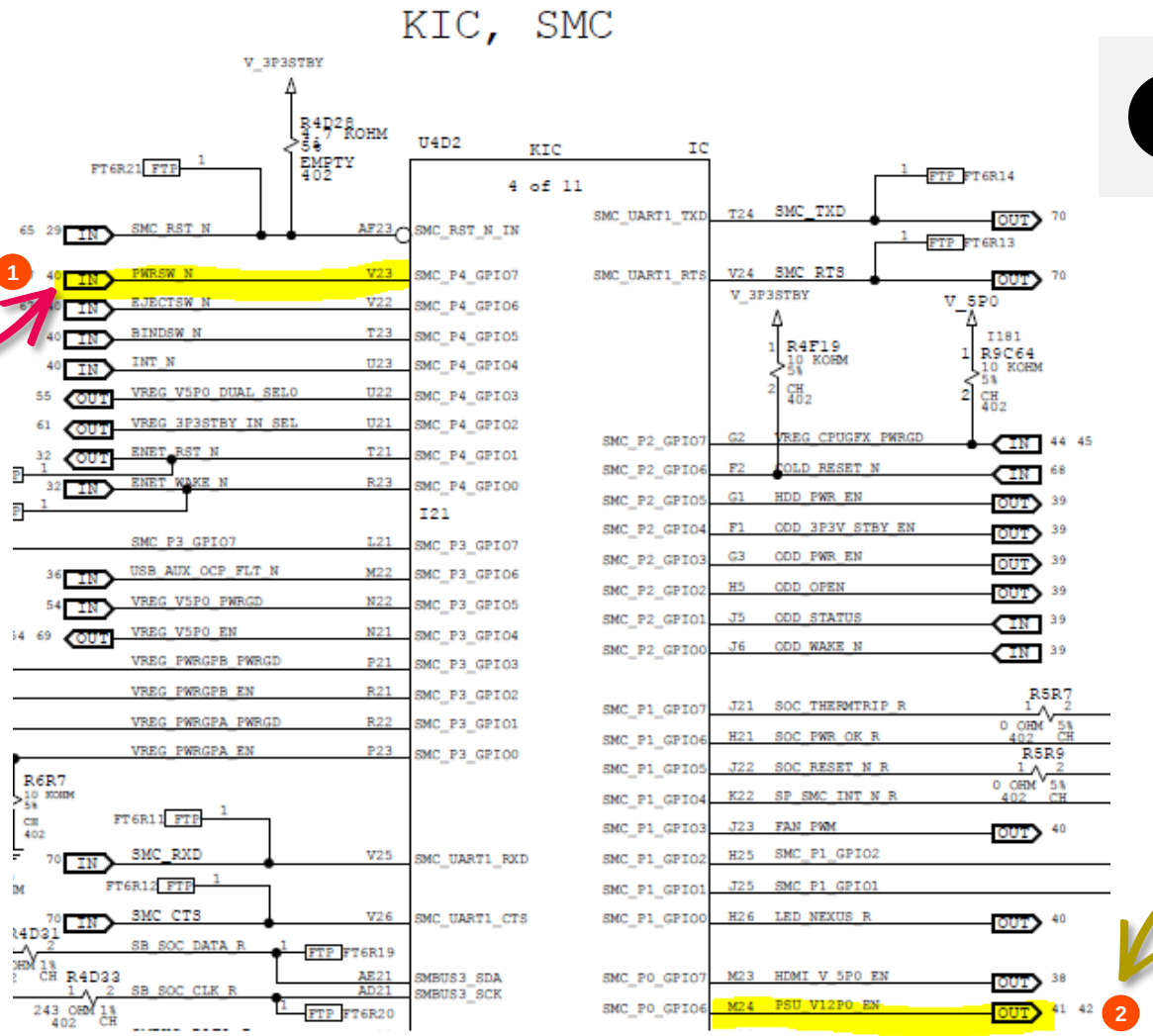
Step 1: Press power button to turn on

When the standby powerrails are present it is time to push the power button on the front panel

1 The PWRSW_N signal will go the the SMC (KIC)



Step 2: Power signal PWRSW_N goes to SMC



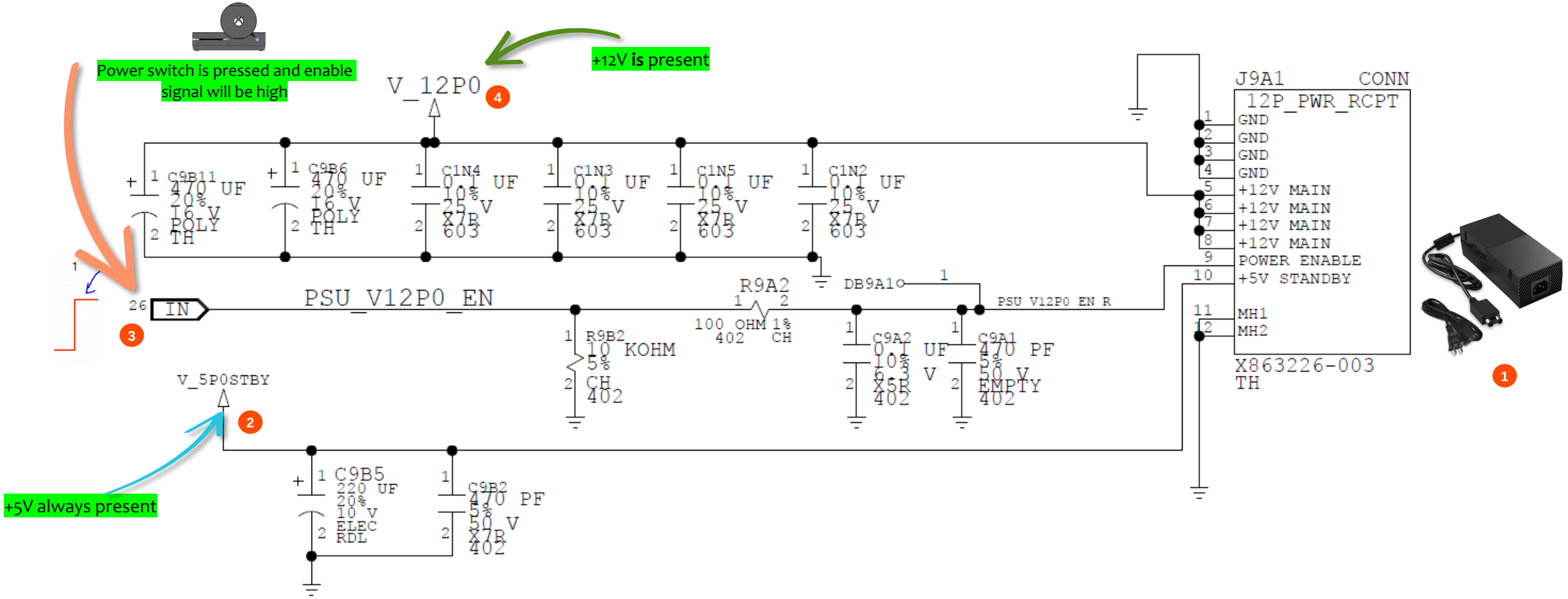
IN: PWRSW_N signal from power button

OUT: signal **PSU_12VP0_EN** will Go to the power connector

2 Signal **PSU_12VP0_EN** will go to the power connector from there the power source will deliver +12V Main

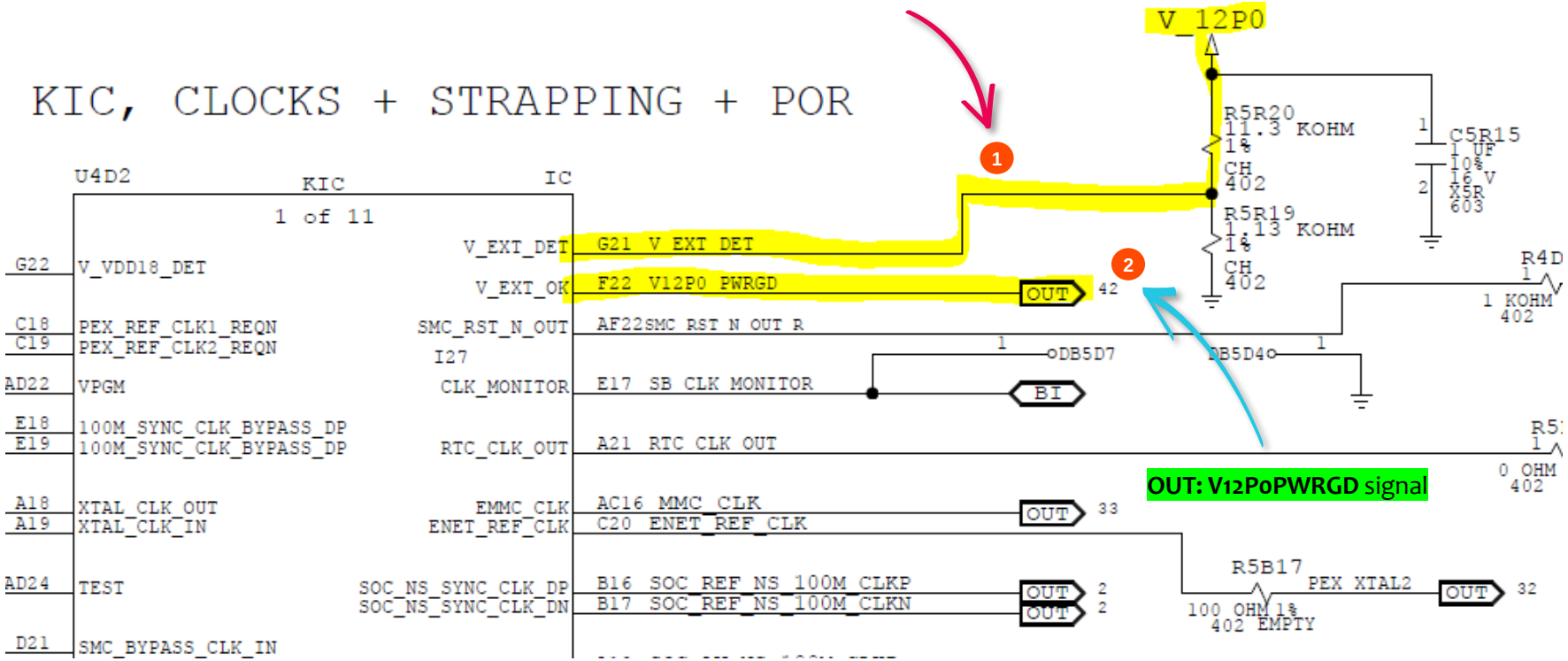
Step 3: Power supply will deliver V_12P0

The signal **PSU_V12P0_EN** will be high and goes to the power supply. The power supply now will deliver **V_12P0** powerrail



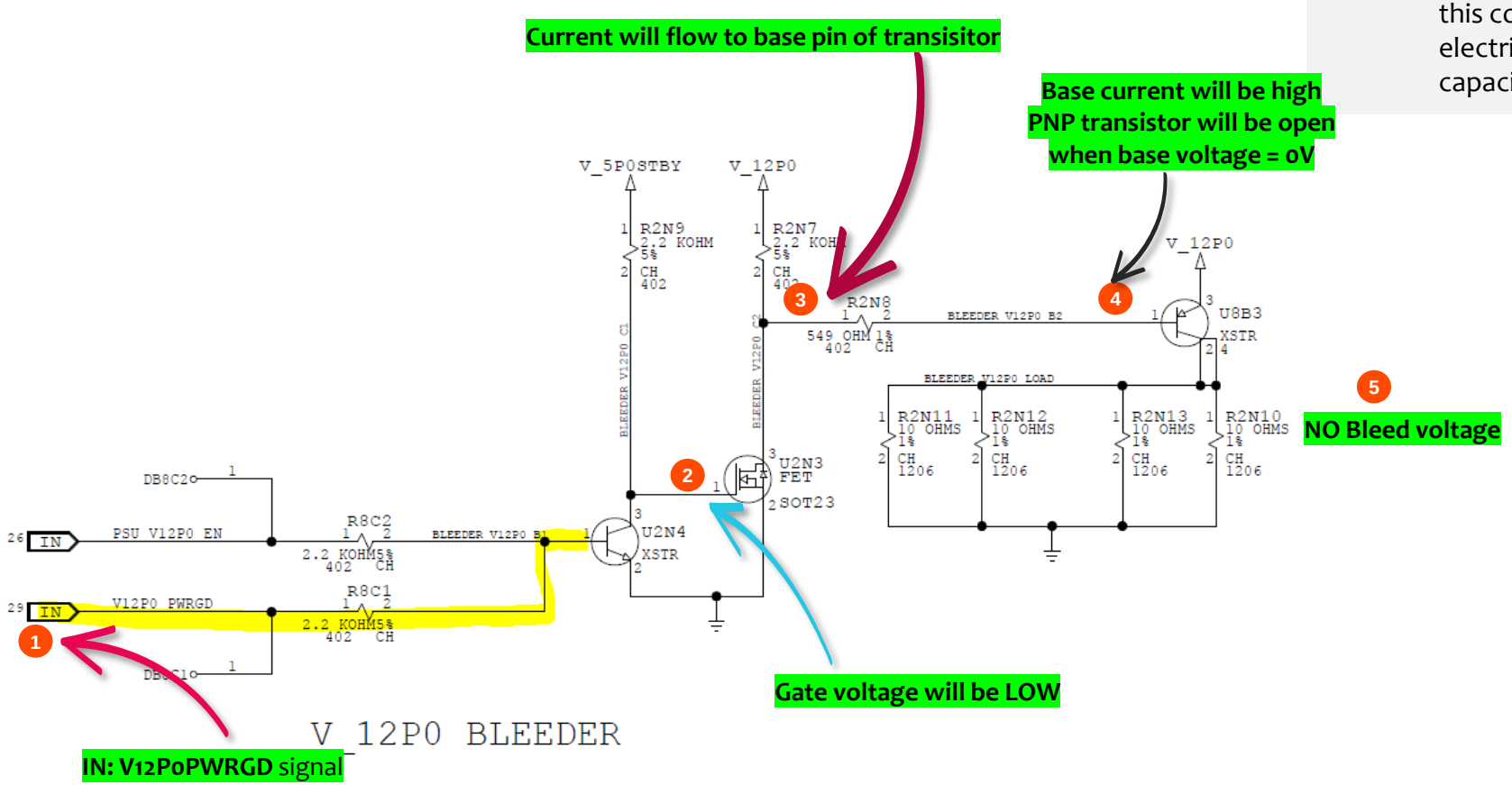
Step 4: Southbridge to detect 12V and send V12P0_PWRGD signal

4 The Southbridge will send a V12P0PWRGD enable signal

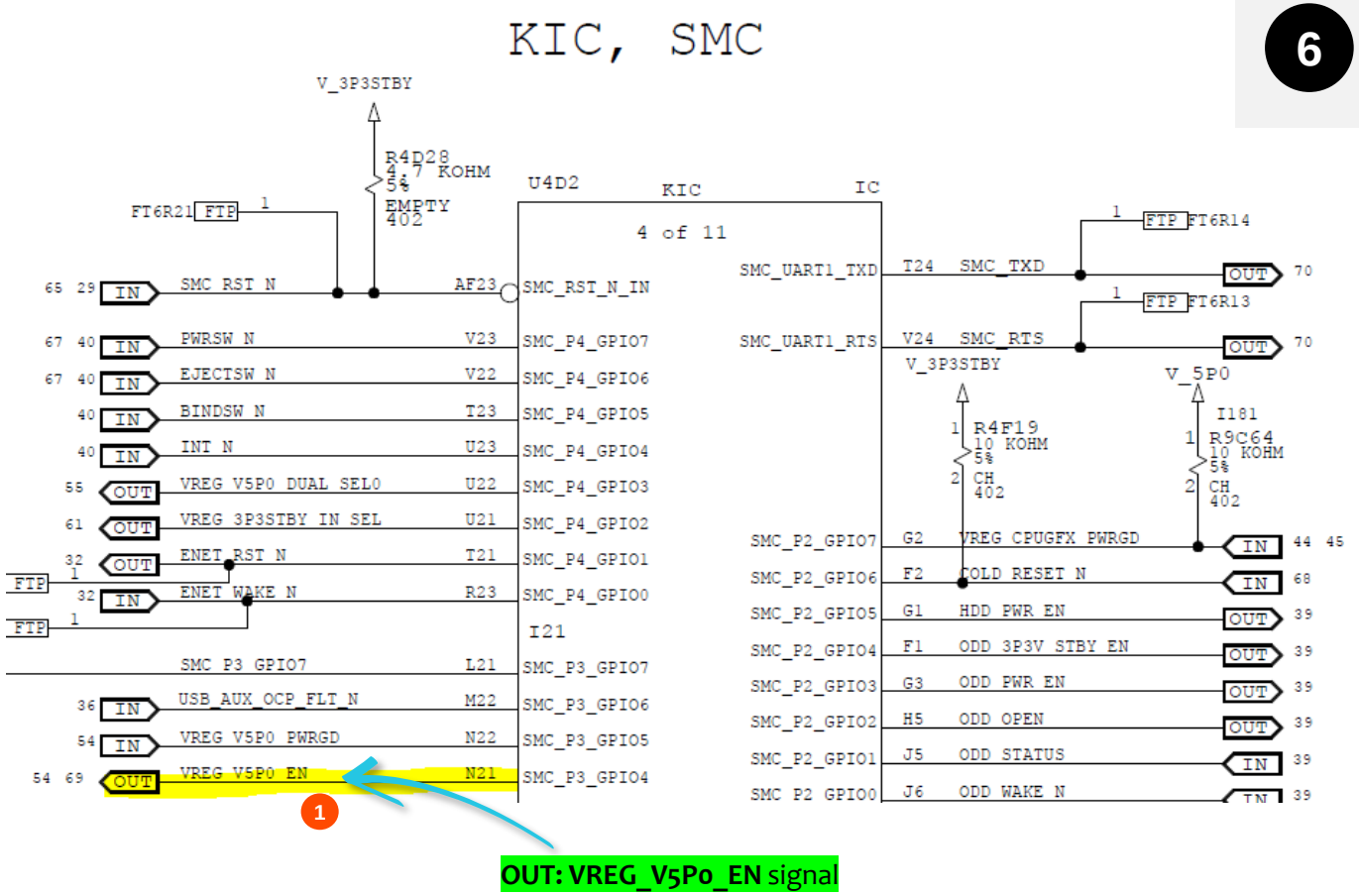


Step 5: Signal V12P0_PWRGD to the +12V bleeder circuit

5 A bleeder resistor is mainly used for safety purposes. It is a type of resistor that is connected to a high voltage power supply circuit in a parallel connection. The purpose of this connection set up to discharge the excess electric charge which is stored in the filter capacitor even after we turn off the device.

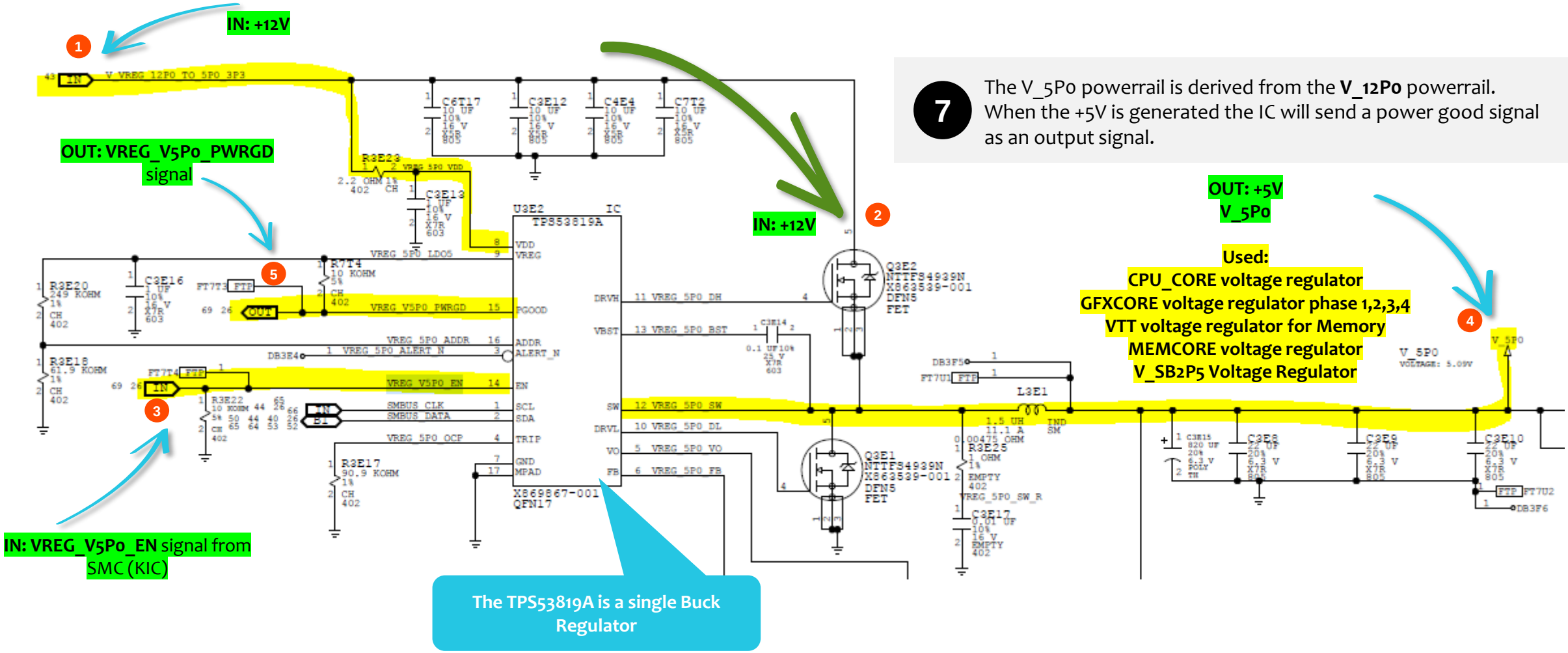


Step 6: SMC send VREG_V5P0_EN signal



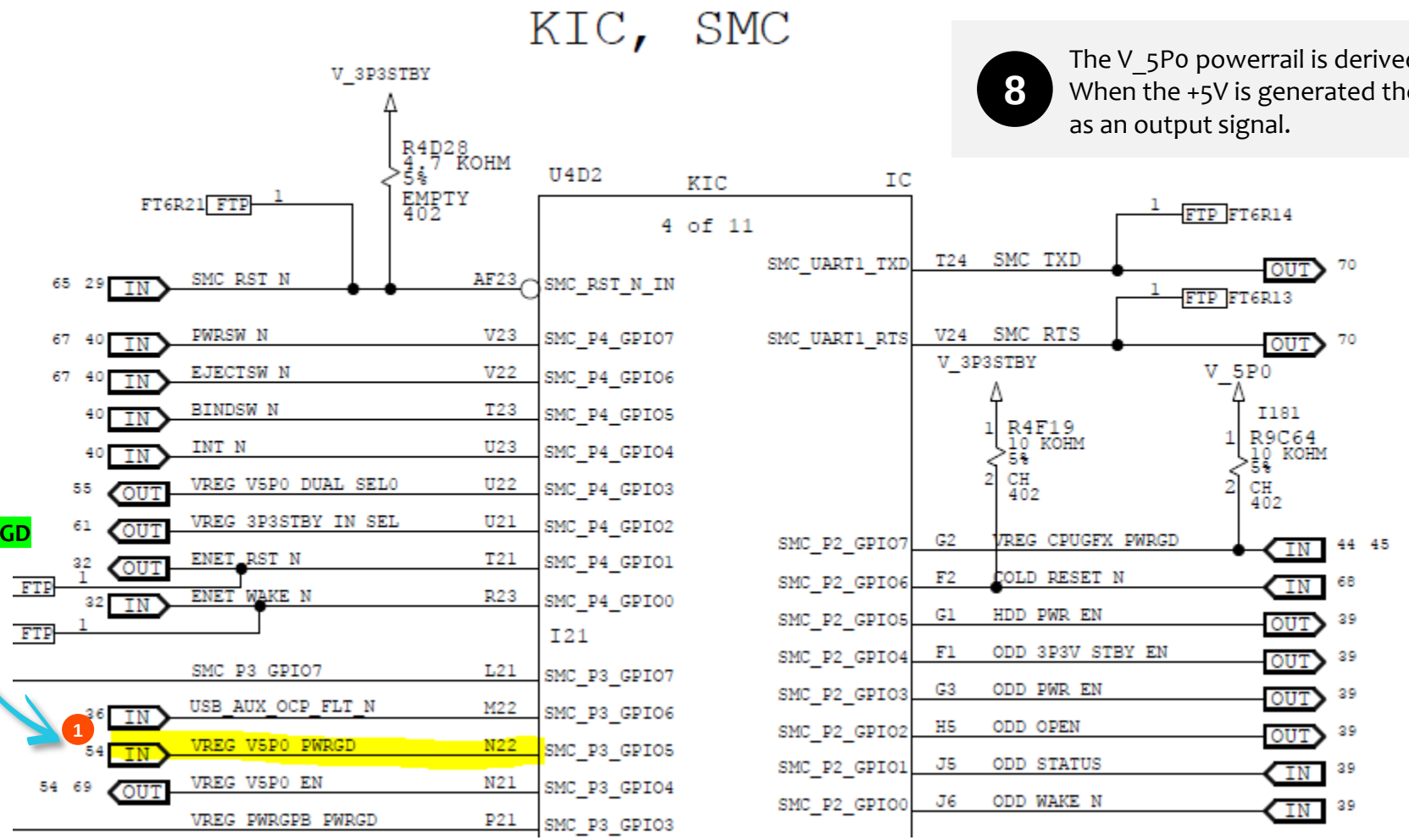
6 This signal is send to a voltage regulator that will convert +12V to +5V

Step 7: Voltage Regulator Module to make +5V



7 The V_5P0 powerrail is derived from the **V_12P0** powerrail. When the +5V is generated the IC will send a power good signal as an output signal.

Step 8: VREG_V5P0_PWRGD signal to SMC

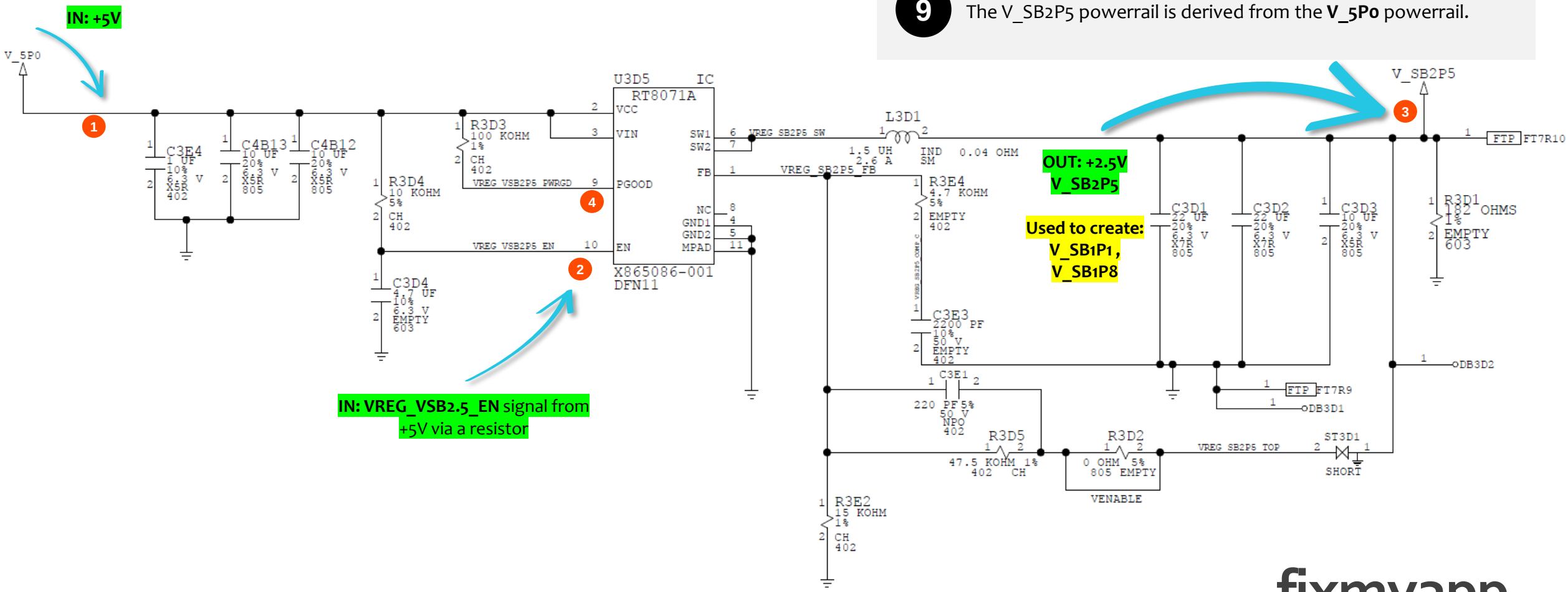


IN: VREG_V5P0_PWRGD signal

8 The V_5P0 powerrail is derived from the **V_12P0** powerrail. When the +5V is generated the IC will send a power good signal as an output signal.

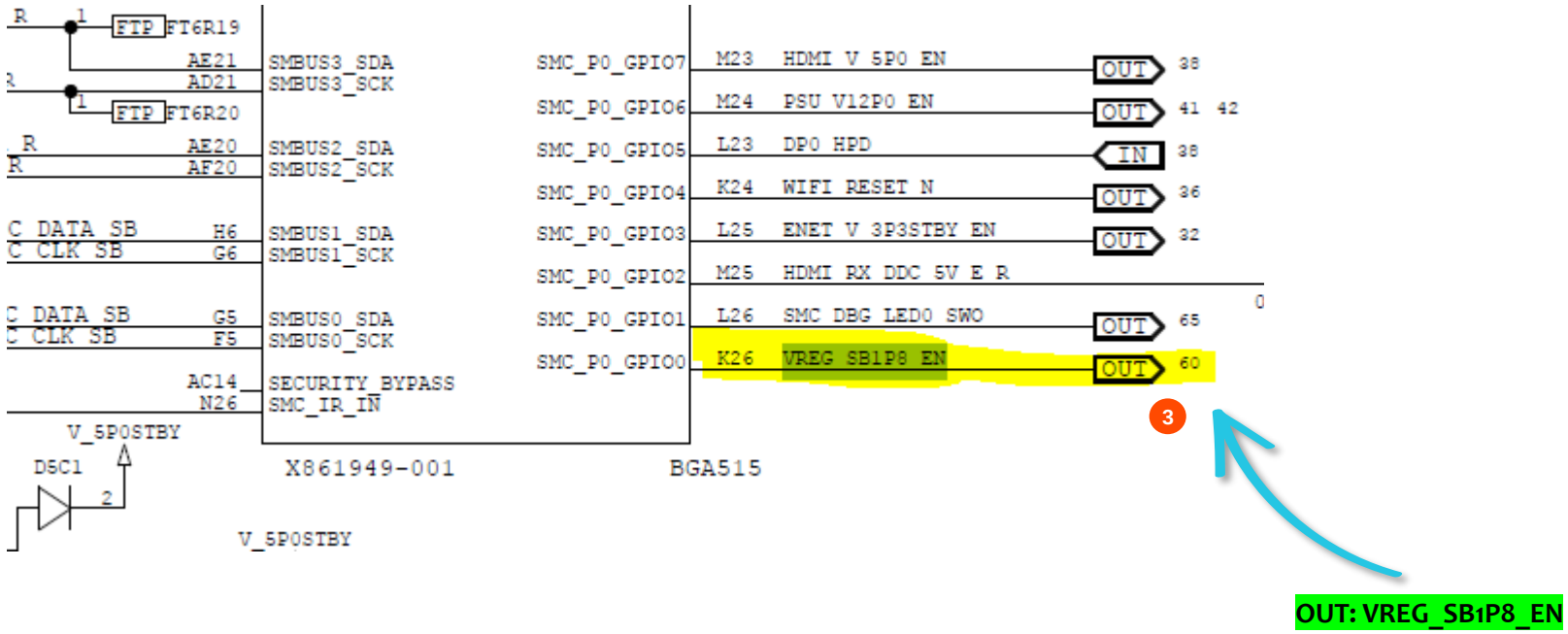
Step 9: Voltage Regulator Module to make +2.5V

9 The V_SB2P5 powerrail is derived from the **V_5P0** powerrail.



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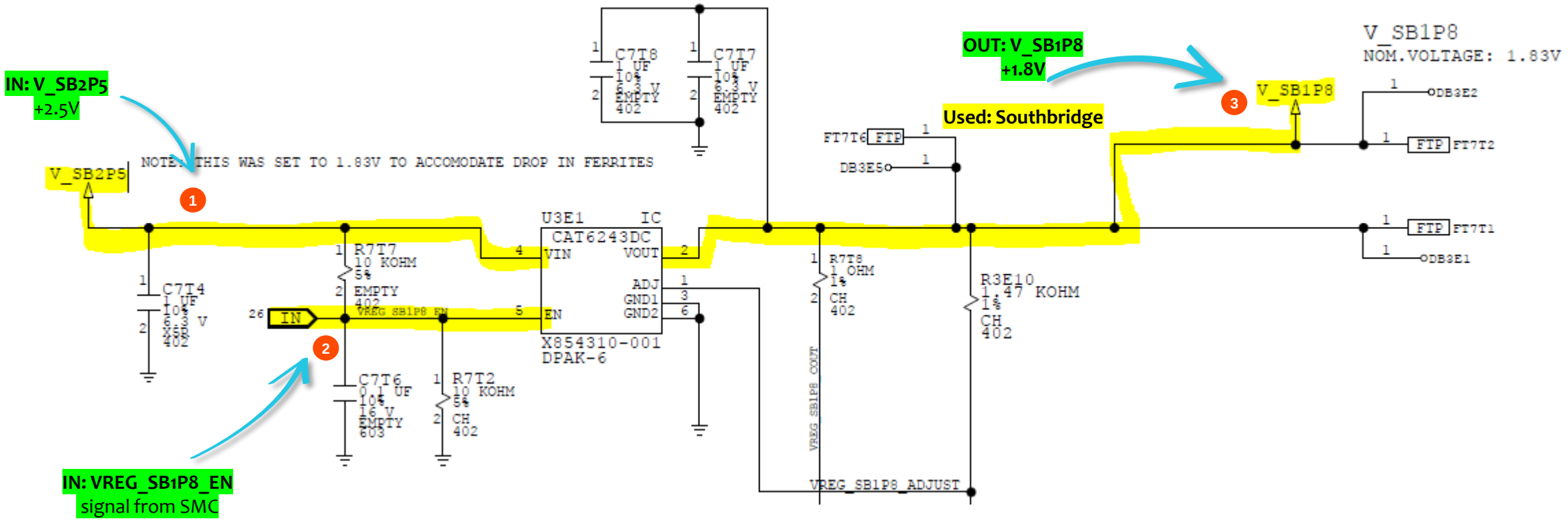
Step 10: SMC (KIC) sends enable signal to VREG to make +1.8V



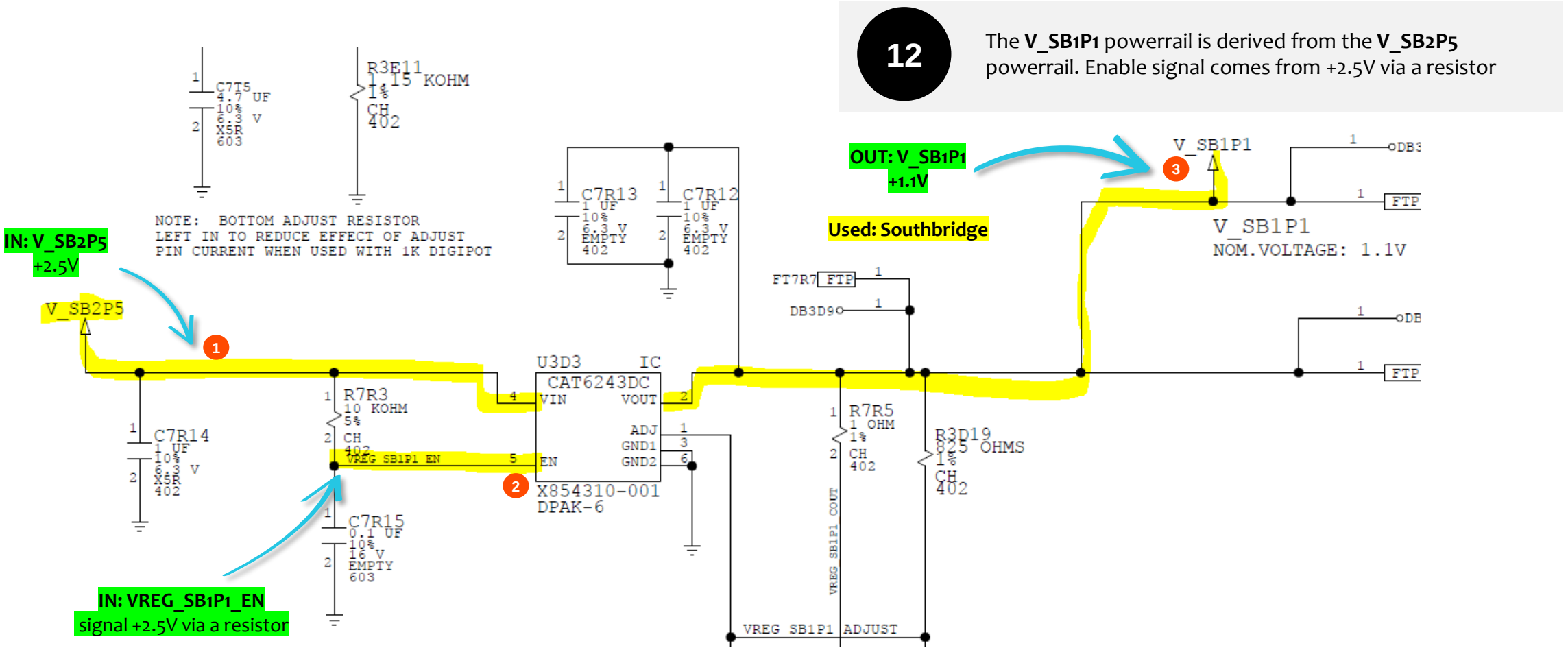
Step 11: Voltage Regulator Module to make +1.8V

11

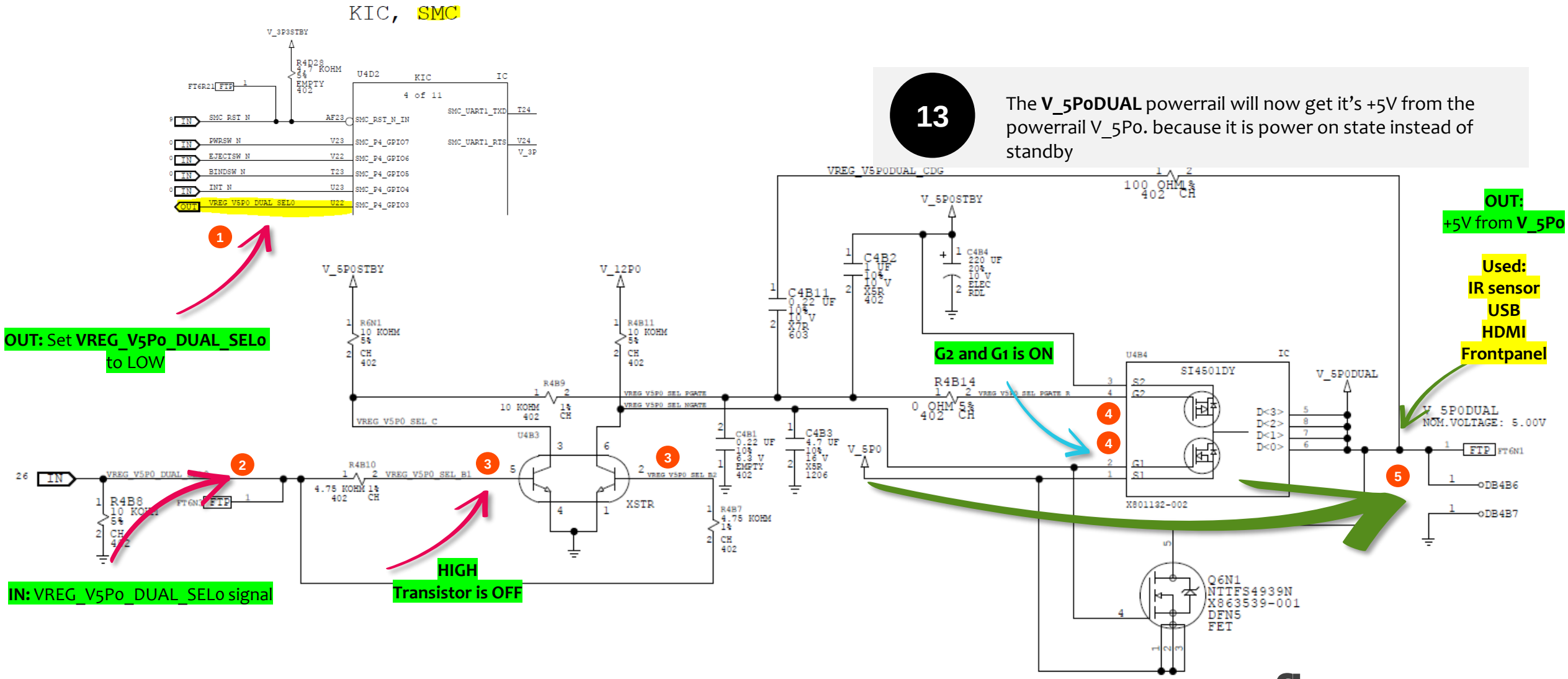
The **V_SB1P8** powerrail is derived from the **V_SB2P5** powerrail. Enable signal comes from the SMC



Step 12: Voltage Regulator Module to make +1.1V

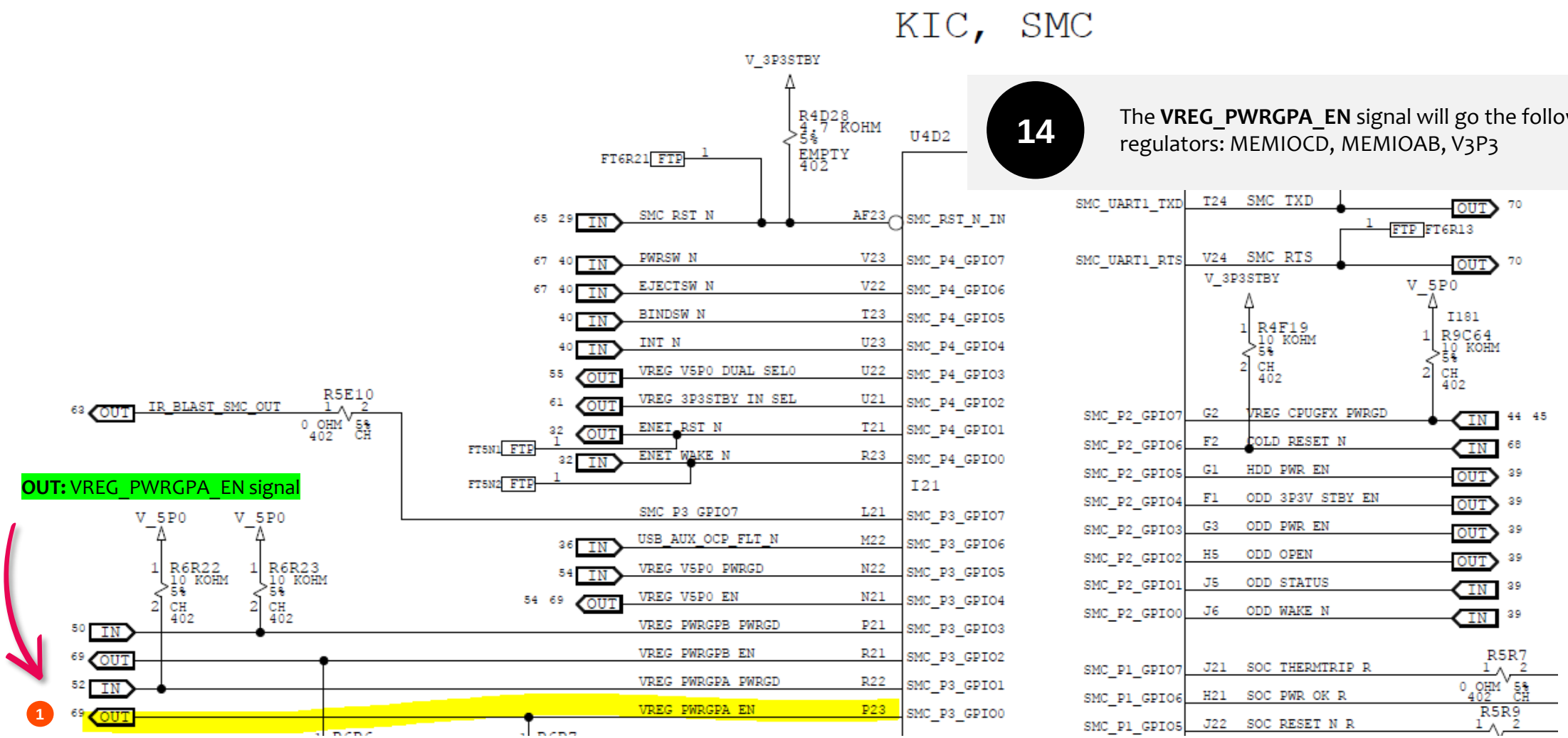


Step 13: Voltage switchers for V_5P0DUAL



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Step 14: SMC sends enable signal VREG_PWRGPA_EN

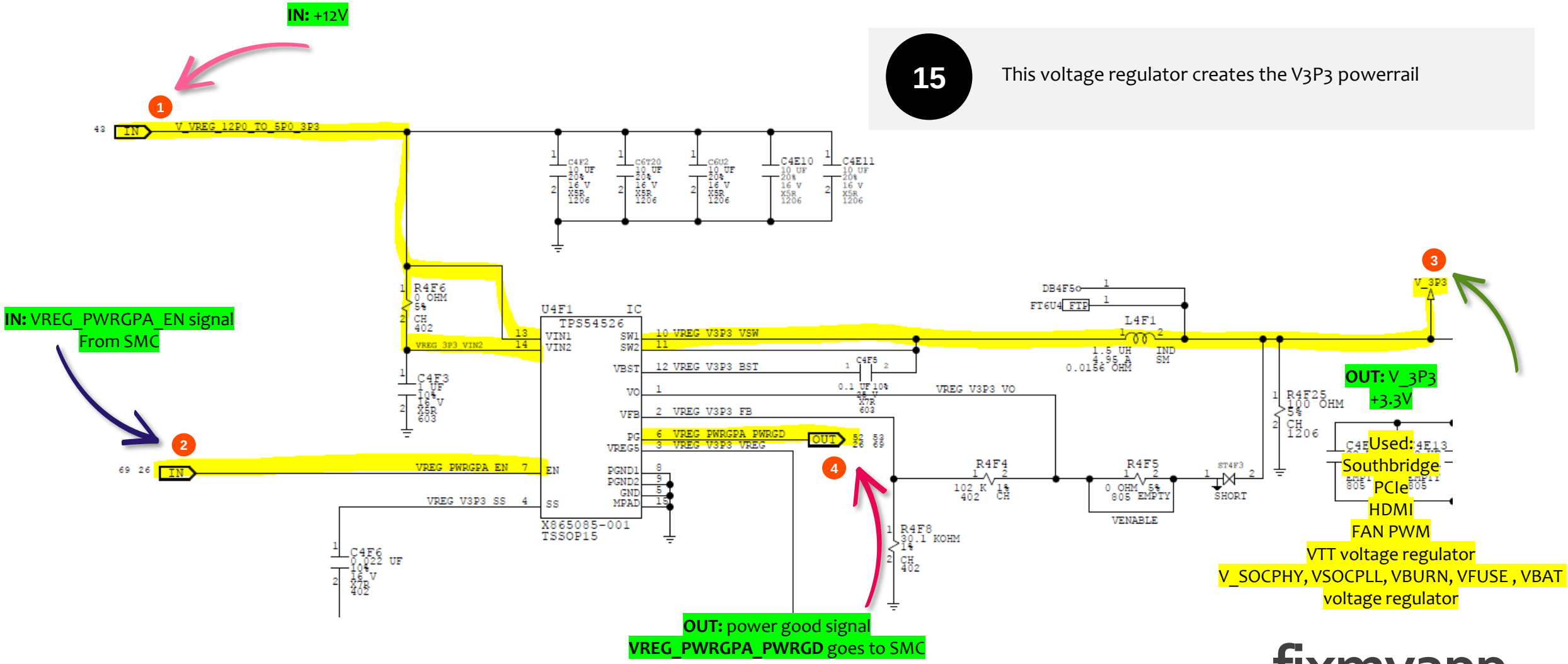


14

The **VREG_PWRGPA_EN** signal will go the following voltage regulators: MEMIOCD, MEMIOAB, V3P3

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Step 15: Voltage regulator will create +3.3V

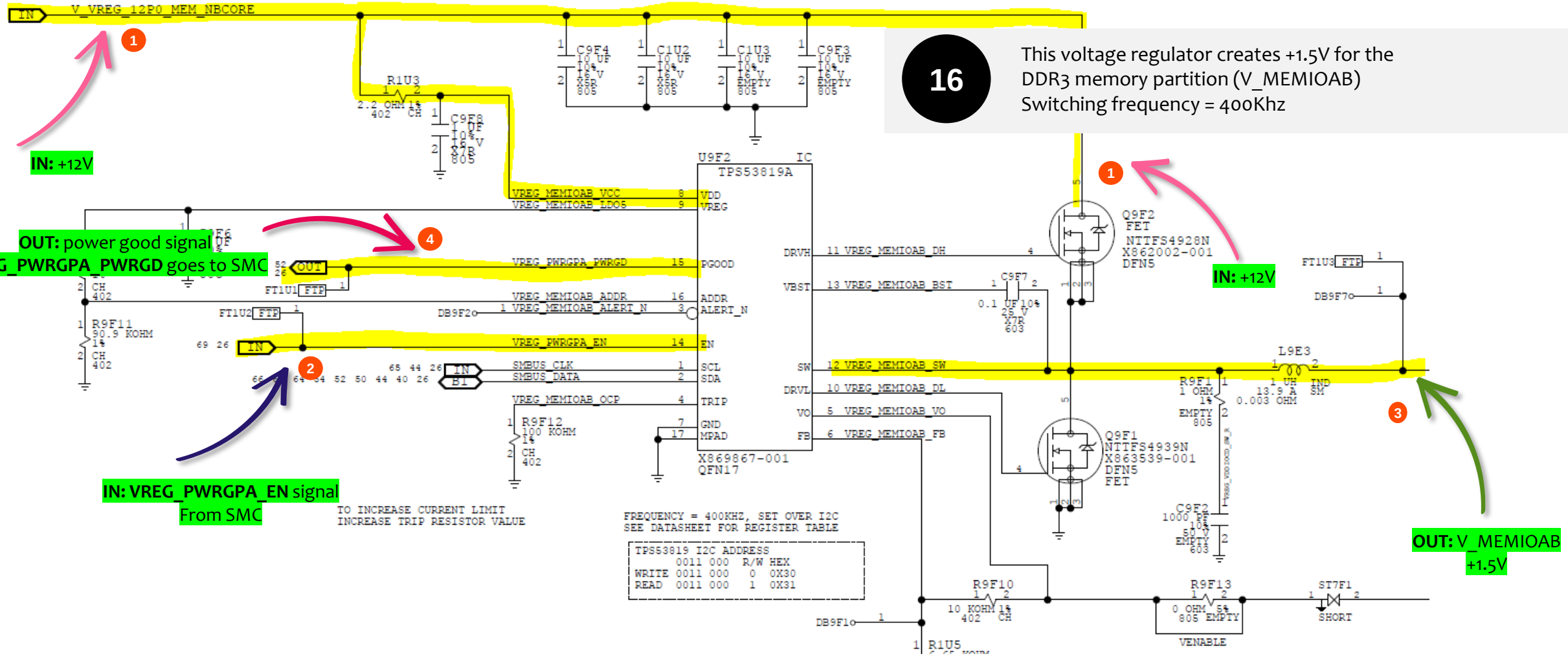


15

This voltage regulator creates the V3P3 powerrail

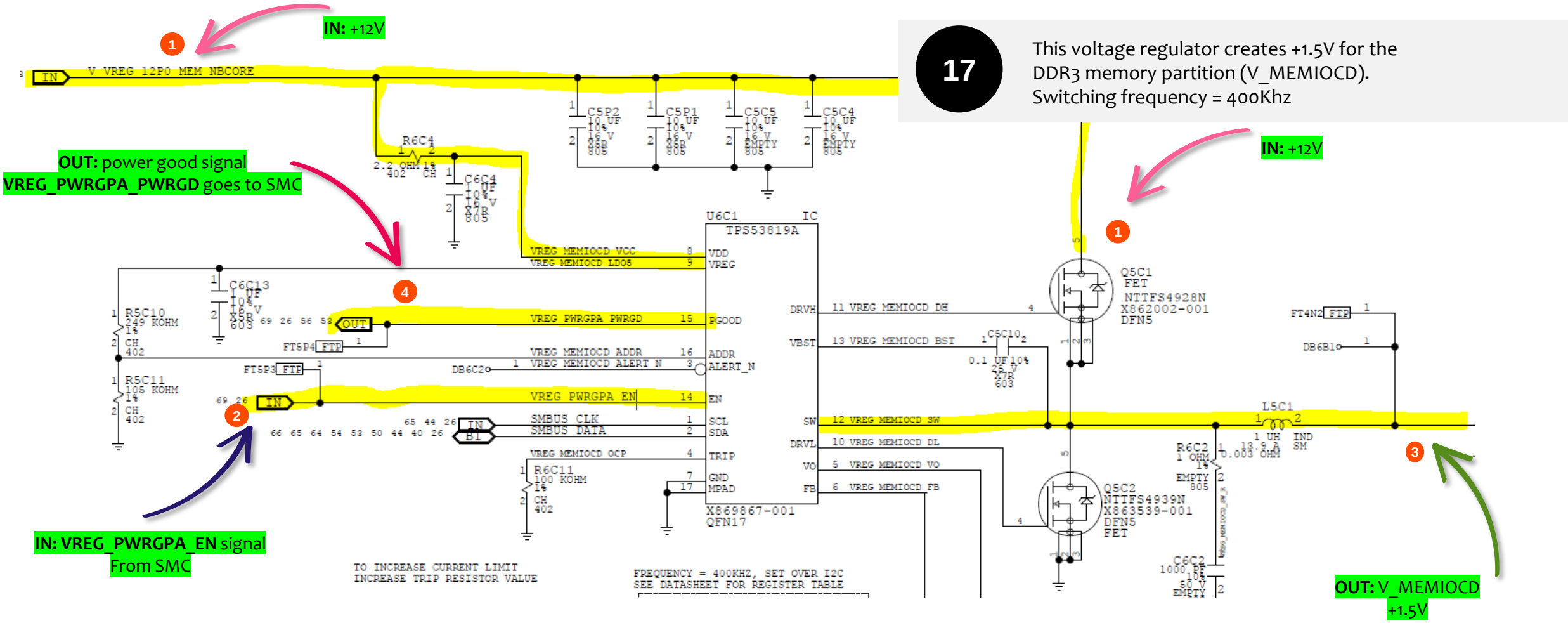
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Step 16: Voltage regulator will create power for the memory partition AB



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Step 17: Voltage regulator will create power for the memory partition CD



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Step 18: Select V_5P0 as source for VREG_3P3STBY_VCC

To power the chip that creates the **V_3P3STBY** voltage the SMC has to send an enable signal to the voltage switcher to determine from wich voltage source will deliver 5V.

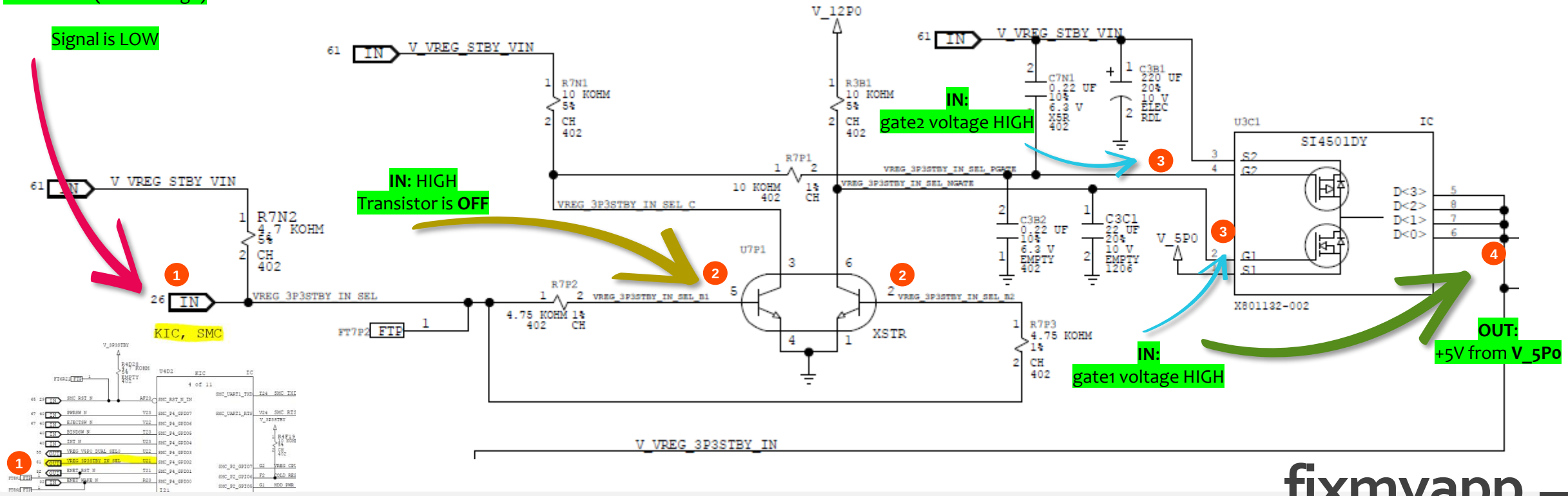
The enable signal **VREG_3P3STBY_IN_SEL** is LOW then the +5V source will be **V_5P0 (power on)**

18

From the **+5V** voltage rail, the 3.3V standby powerrail is created called **V_3P3STBY (+3.3V)**

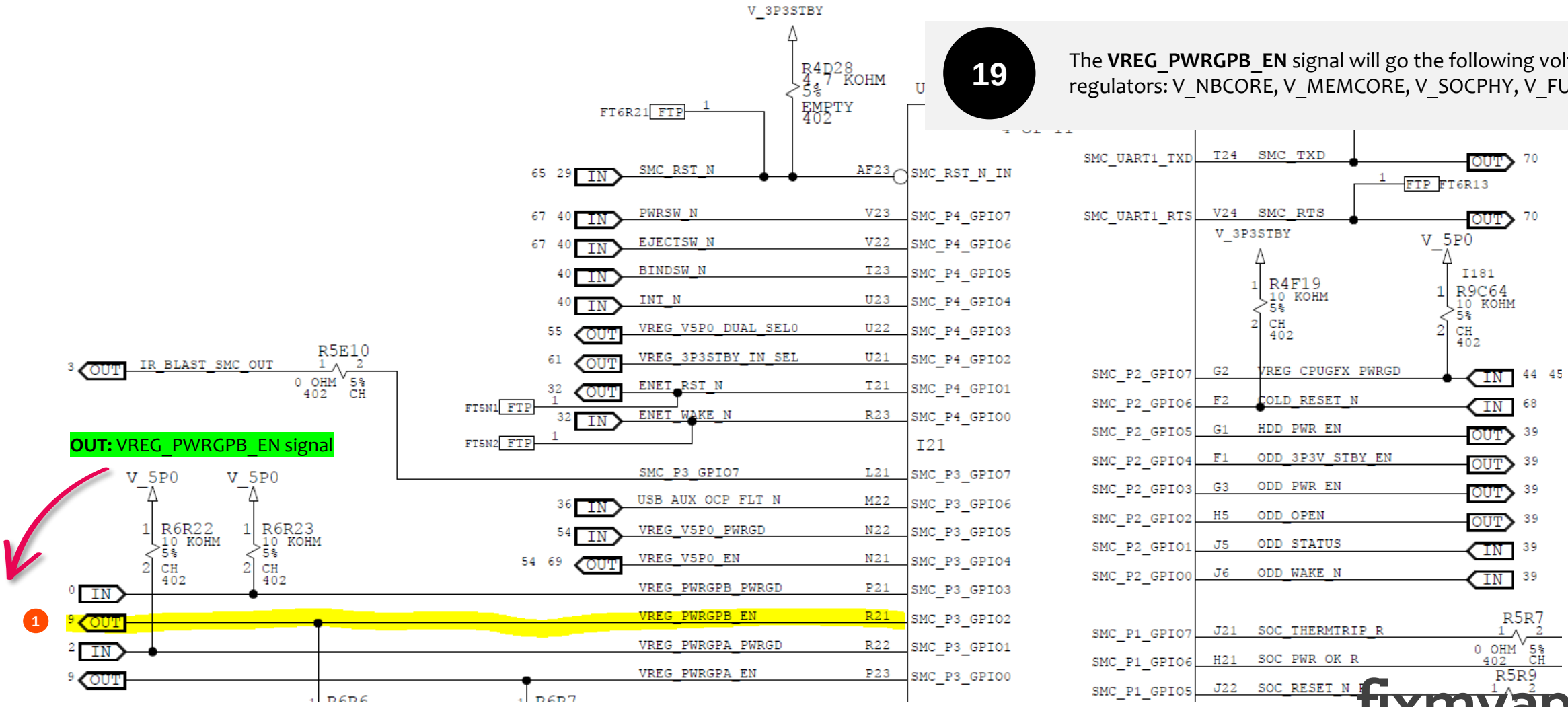
IN: the signal comes from the SMC IC (Southbridge)

Signal is LOW



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Step 19: SMC sends enable signal VREG_PWRGPB_EN



19

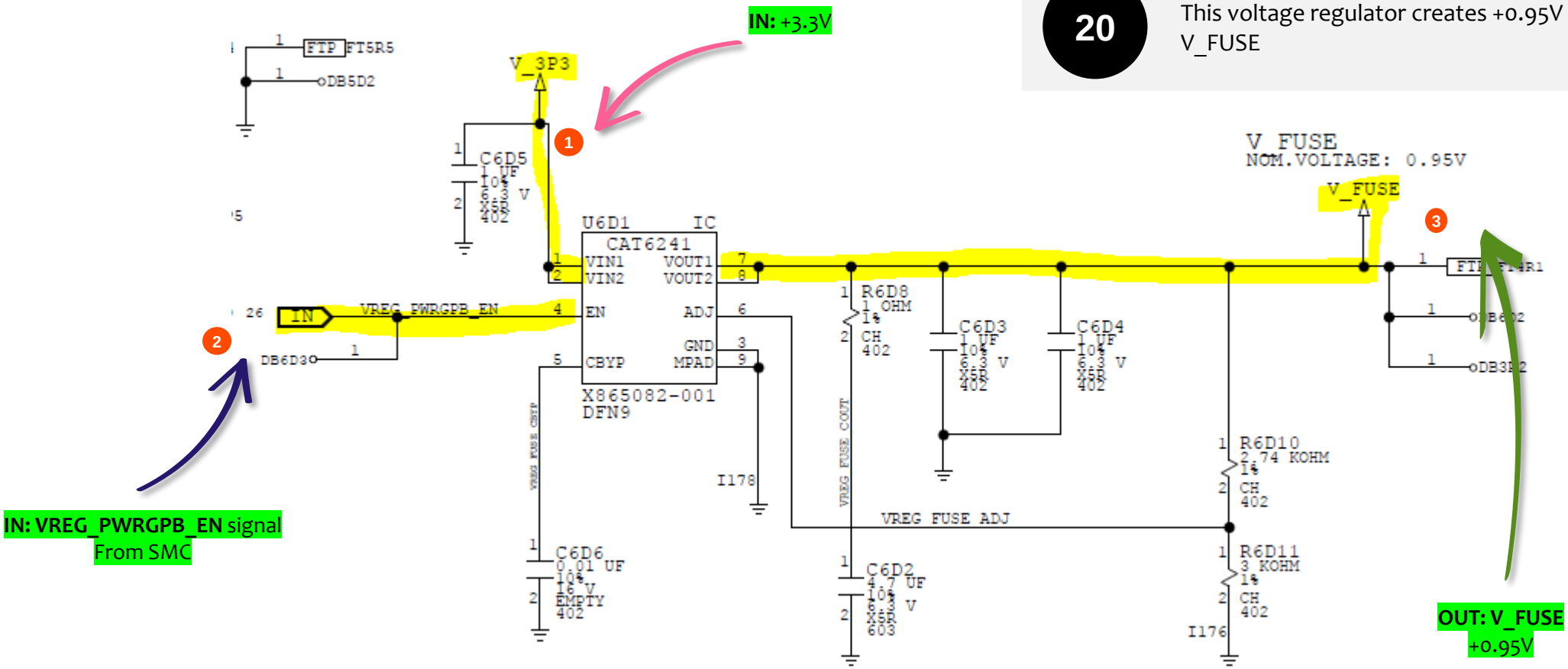
The **VREG_PWRGPB_EN** signal will go the following voltage regulators: V_NBCORE, V_MEMCORE, V_SOCPHY, V_FUSE

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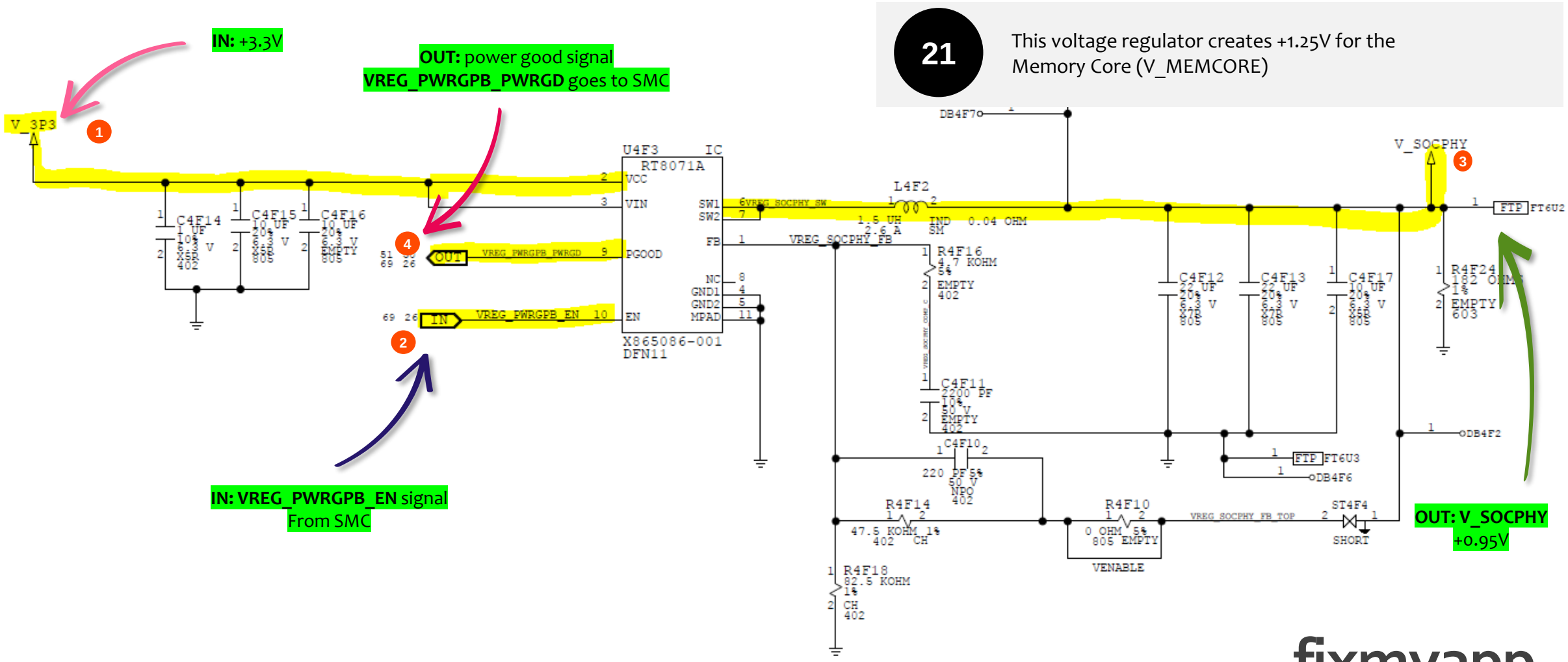
Step 20: Voltage regulator will create power +0.95V (V_FUSE)

20

This voltage regulator creates +0.95V for the V_FUSE

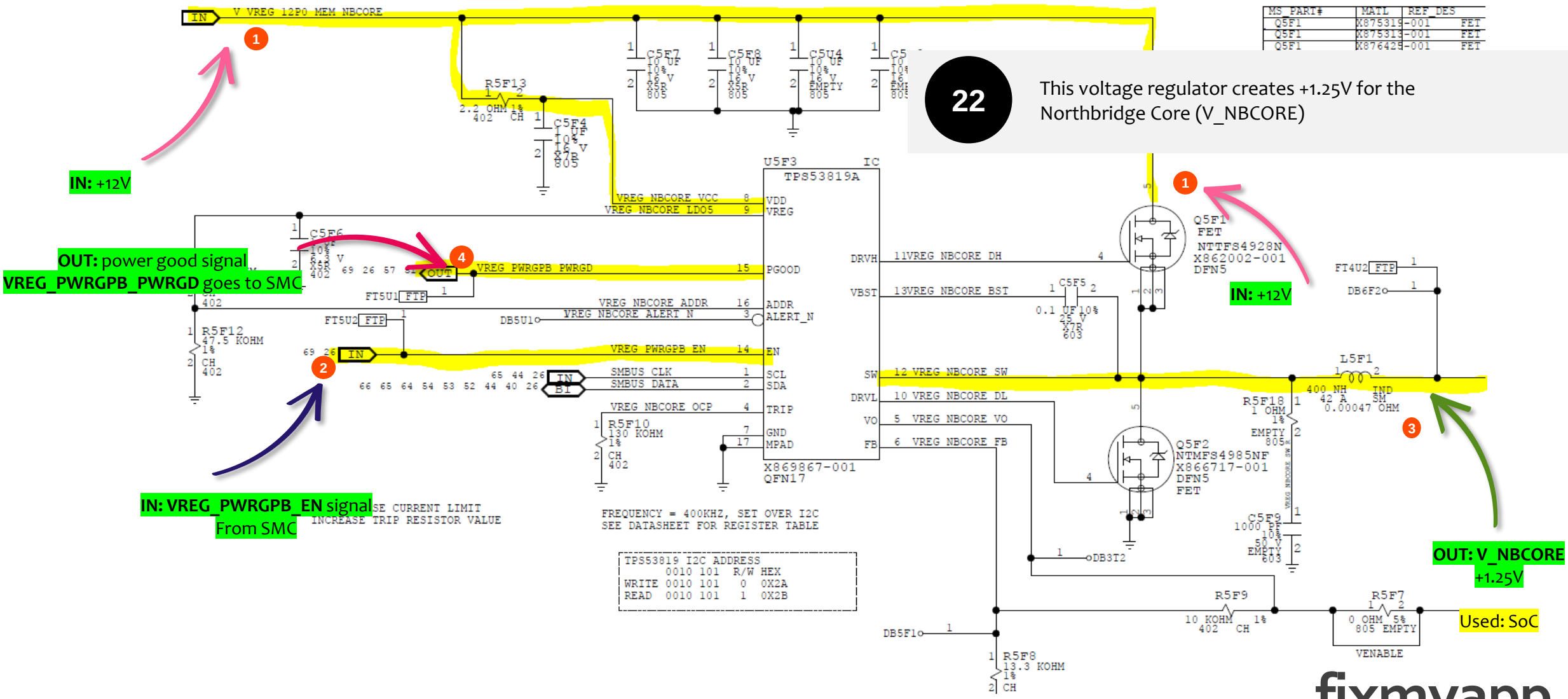


Step 21: Voltage regulator will create power +0.95V (V_SOCPHY)



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Step 22: Voltage regulator will create power +1.25V (V_NBCORE)

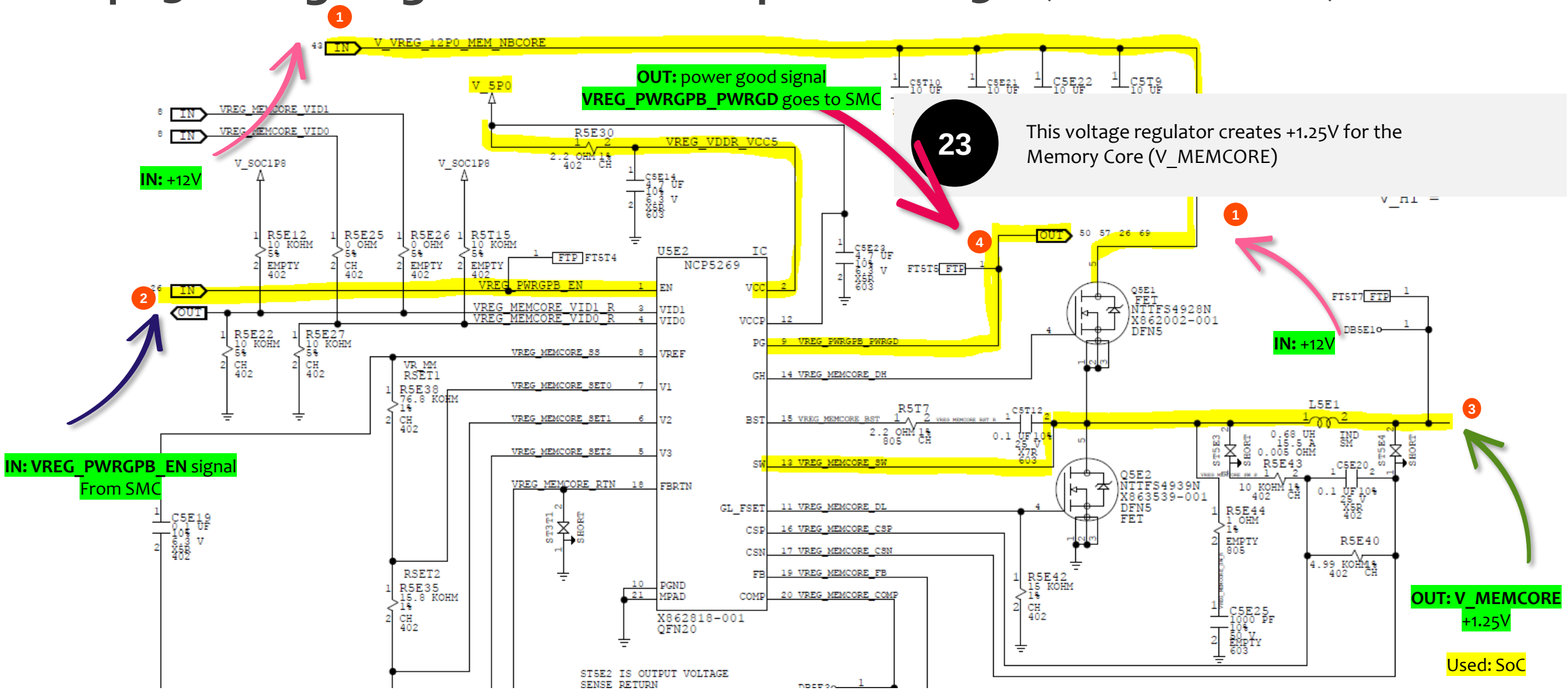


MS_PART#	MATL	REF_DES	
Q5F1	X875319-001		FET
Q5F1	X875313-001		FET
Q5F1	X876429-001		FET

22

This voltage regulator creates +1.25V for the Northbridge Core (V_NBCORE)

Step 23: Voltage regulator will create power +1.25V (V_MEMCORE)



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NCP4204 – What is it? – No datasheet

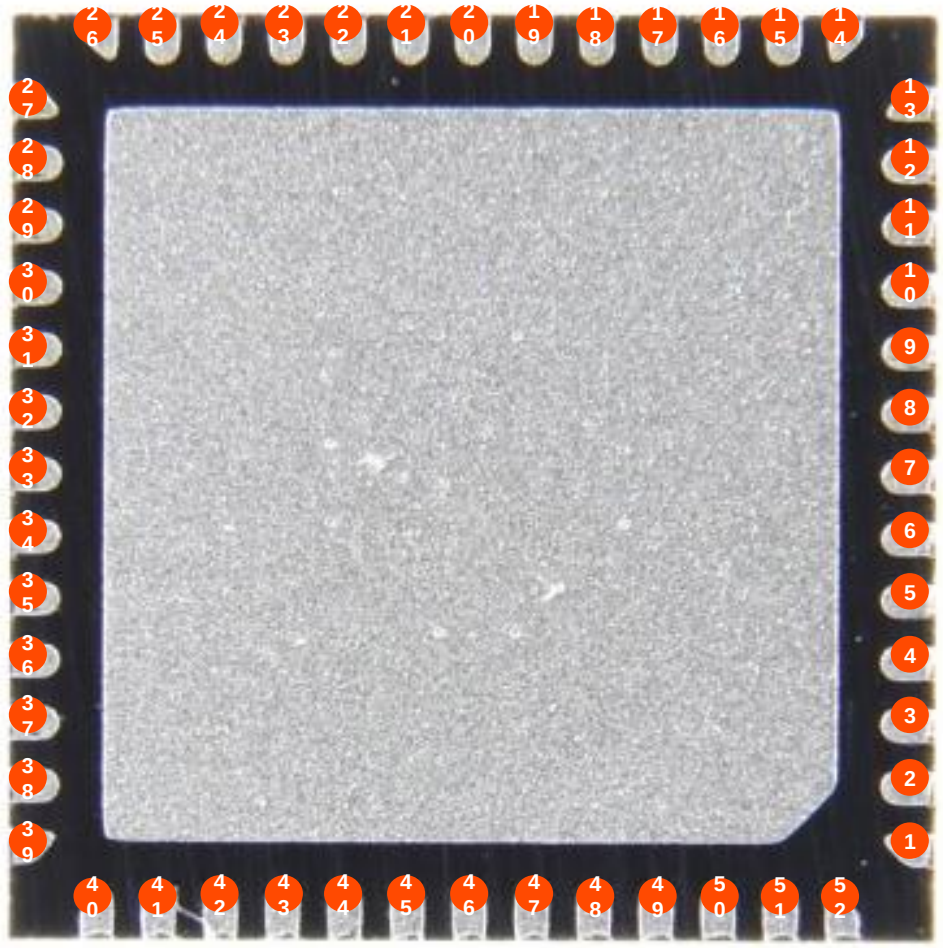


The NCP4204 is a specific voltage regulator IC (integrated circuit) manufactured by ON Semiconductor. It is designed for use in power supply applications and features a maximum output current of **5A**.

The NCP4204 has:

1 output 1 phase for the CPU Core
1 output 4 phase for the GFX Core

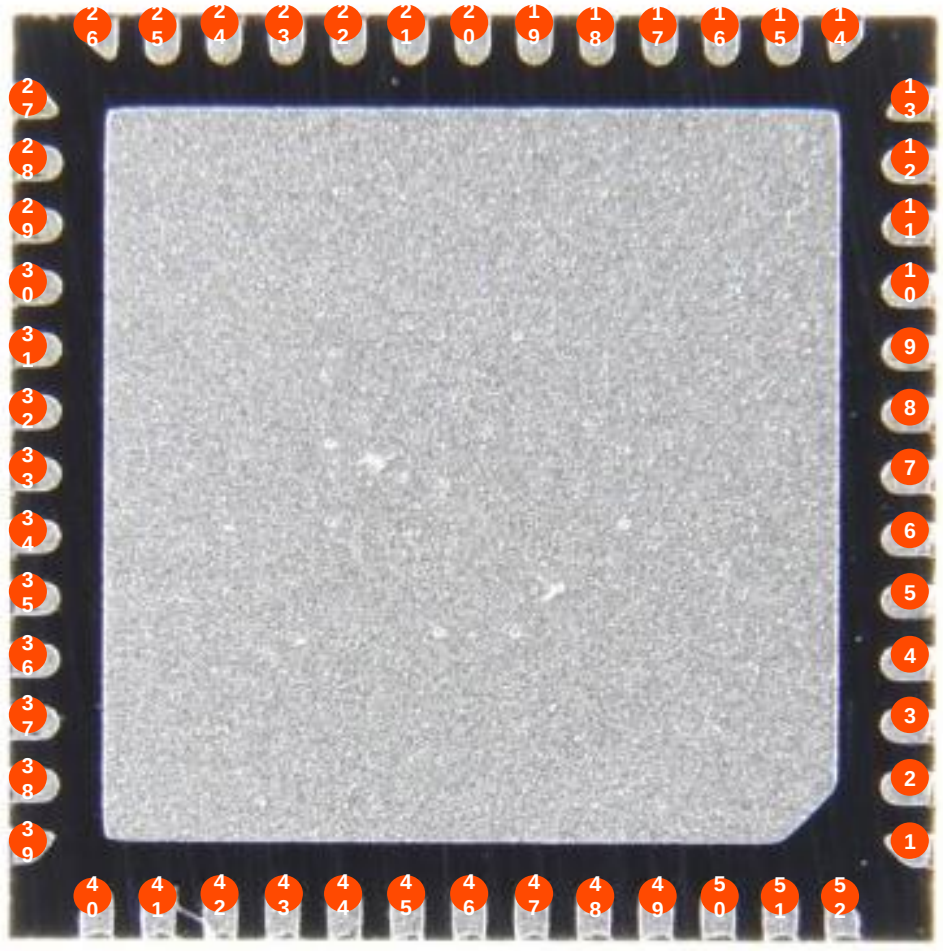
NCP4204 – pinout



1 = BI: SVD – SoC <> NCP
2 = OUT: SVT - To SoC
3 = IN: SVC – From Soc
4 = IN: VDDIO – From V_SOC1P8
5 = OUT: SCL (clock)
6 = BI: SDA (I2C)
7 = OUT: VDD_PWRGD – To Southbridge
8 = OUT: VDDNB_PWRGD – To Southbridge
9 = IN: EN – From Southbridge
10 = IN: EN_NB – From Southbridge
11 = IN: VCC – From V_5Po
12 = IN: VRMP – From +12V
13 = OUT: OCP_L
14 = IN: VSS – From SoC
15 = IN: VDD – From SoC
16 = IN: FB – From pin 17
17 = OUT: DIFF - To Pin 16
18 = IN: TRBST
19 = IN: COMP
20 = IN: ILIM
21 = IN: DROOP
22 = IN: CSCOMP
23 = OUT: IOUT
24 = IN: CSSUM – total current sensing
25 = IN: CSP1 – From ouput V_CPUCORE
26 = IN: CSN1 – From ouput V_CPUCORE
27 = OUT: PWM1 – To CPUCORE voltage regulator IC
28 = OUT: PWM4NB – To GFXCORE voltage regulator IC phase 4
29 = OUT: PWM2NB – To GFXCORE voltage regulator IC phase 2
30 = OUT: PWM3NB – To GFXCORE voltage regulator IC phase 3
31 = OUT: PWM1NB – To GFXCORE voltage regulator IC phase 1
32 = OUT: DRON – To GFXCORE enable signal phase 1,2,3,4
33 = IN: CSP1NB – From GFXCORE output phase 1
34 = IN: CSN1NB – From From GFXCORE output phase 1
35 = IN: CSP3NB – From GFXCORE output phase 3
36 = IN: CSN3NB – From GFXCORE output phase 3
37 = IN: CSP2NB – From GFXCORE output phase 2
38 = IN: CSN2NB – From GFXCORE output phase 2
39 = IN: CSN4NB – From GFXCORE output phase 4
40 = IN: CSP4NB – From GFXCORE output phase 4
41 = IN: CSREFNB – From PIN 41
42 = IOUTNB
43 = IN: CSSUMNB - From GFXCORE output phase 1,2,3,4
44 = IN: CSCOMPNB
45 = IN: DROOPNB
46 = IN: ILMNB
47 = IN: COMPNB
48 = IN: FBNB
49 = IN: TRBSTNB
50 = OUT: DIFFNB – Goes to PIN 47, 48, 49
51 = IN: VDDNB – From SoC
52 = IN: PWROK – Comes from Southbridge

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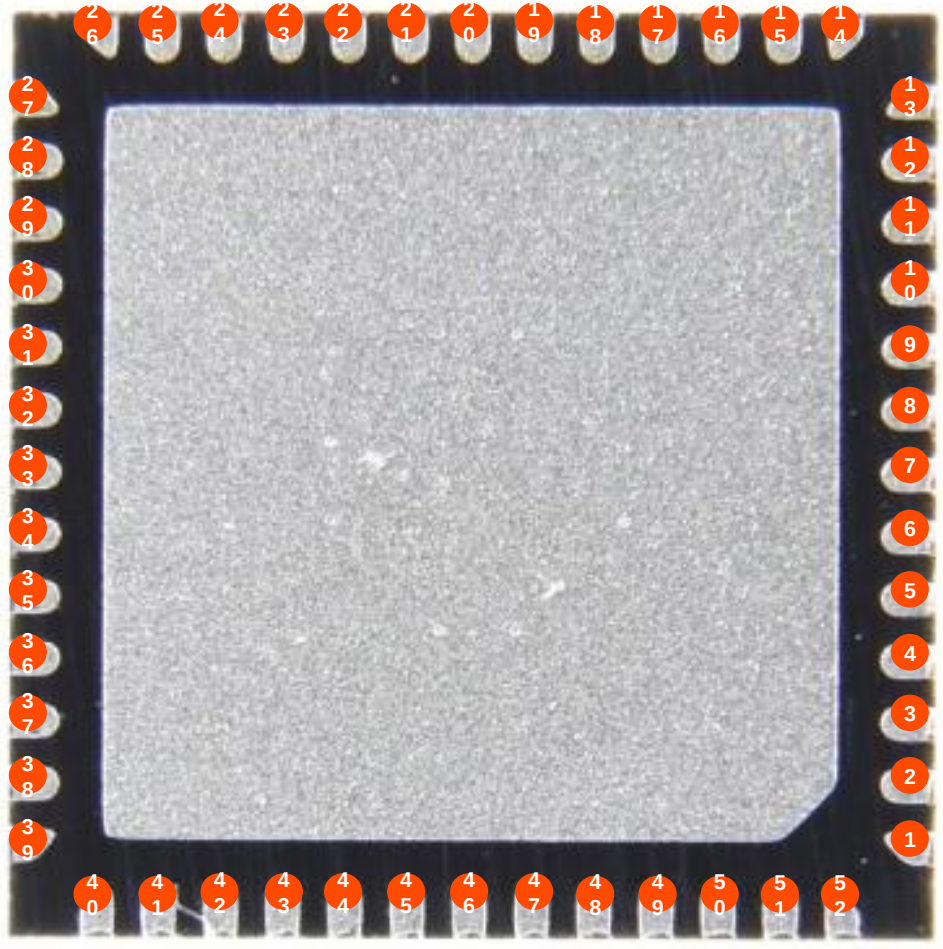
NCP4204 – CPUCORE signals



1 = BI: SVD – SoC <> NCP
2 = OUT: SVT - To SoC
3 = IN: SVC – From Soc
4 = IN: VDDIO – From V_SOC1P8
5 = OUT: SCL (clock)
6 = BI: SDA (I2C)
7 = OUT: VDD_PWRGD – To Southbridge
8 = OUT: VDDNB_PWRGD – To Southbridge
9 = IN: EN – From Southbridge
10 = IN: EN_NB – From Southbridge
11 = IN: VCC – From V_5P0
12 = IN: VRMP – From +12V
13 = OUT: OCP_L
14 = IN: VSS – From SoC
15 = IN: VDD – From SoC
16 = IN: FB – From pin 17
17 = OUT: DIFF - To Pin 16
18 = IN: TRBST
19 = IN: COMP
20 = IN: ILIM
21 = IN: DROOP
22 = IN: CSCOMP
23 = OUT: IOUT
24 = IN: CSSUM – total current sensing
25 = IN: CSP1 – From ouput V_CPUCORE
26 = IN: CSN1 – From ouput V_CPUCORE
27 = OUT: PWM1 – To CPUCORE voltage regulator IC
28 = OUT: PWM4NB – To GFXCORE voltage regulator IC phase 4

29 = OUT: PWM2NB – To GFXCORE voltage regulator IC phase 2
30 = OUT: PWM3NB – To GFXCORE voltage regulator IC phase 3
31 = OUT: PWM1NB – To GFXCORE voltage regulator IC phase 1
32 = OUT: DRON – To CPUCORE and GFXCORE enable signal
33 = IN: CSP1NB – From GFXCORE output phase 1
34 = IN: CSN1NB – From From GFXCORE output phase 1
35 = IN: CSP3NB – From GFXCORE output phase 3
36 = IN: CSN3NB – From GFXCORE output phase 3
37 = IN: CSP2NB – From GFXCORE output phase 2
38 = IN: CSN2NB – From GFXCORE output phase 2
39 = IN: CSN4NB – From GFXCORE output phase 4
40 = IN: CSP4NB – From GFXCORE output phase 4
41 = IN: CSREFNB – From PIN 41
42 = IOUTNB
43 = IN: CSSUMNB - From GFXCORE output phase 1,2,3,4
44 = IN: CSCOMPNB
45 = IN: DROOPNB
46 = IN: ILMNB
47 = IN: COMPNB
48 = IN: FBNB
49 = IN: TRBSTNB
50 = OUT: DIFFNB – Goes to PIN 47, 48, 49
51 = IN: VDDNB – From SoC
52 = IN: PWROK – Comes from Southbridge

NCP4204 – GFXCORE signals



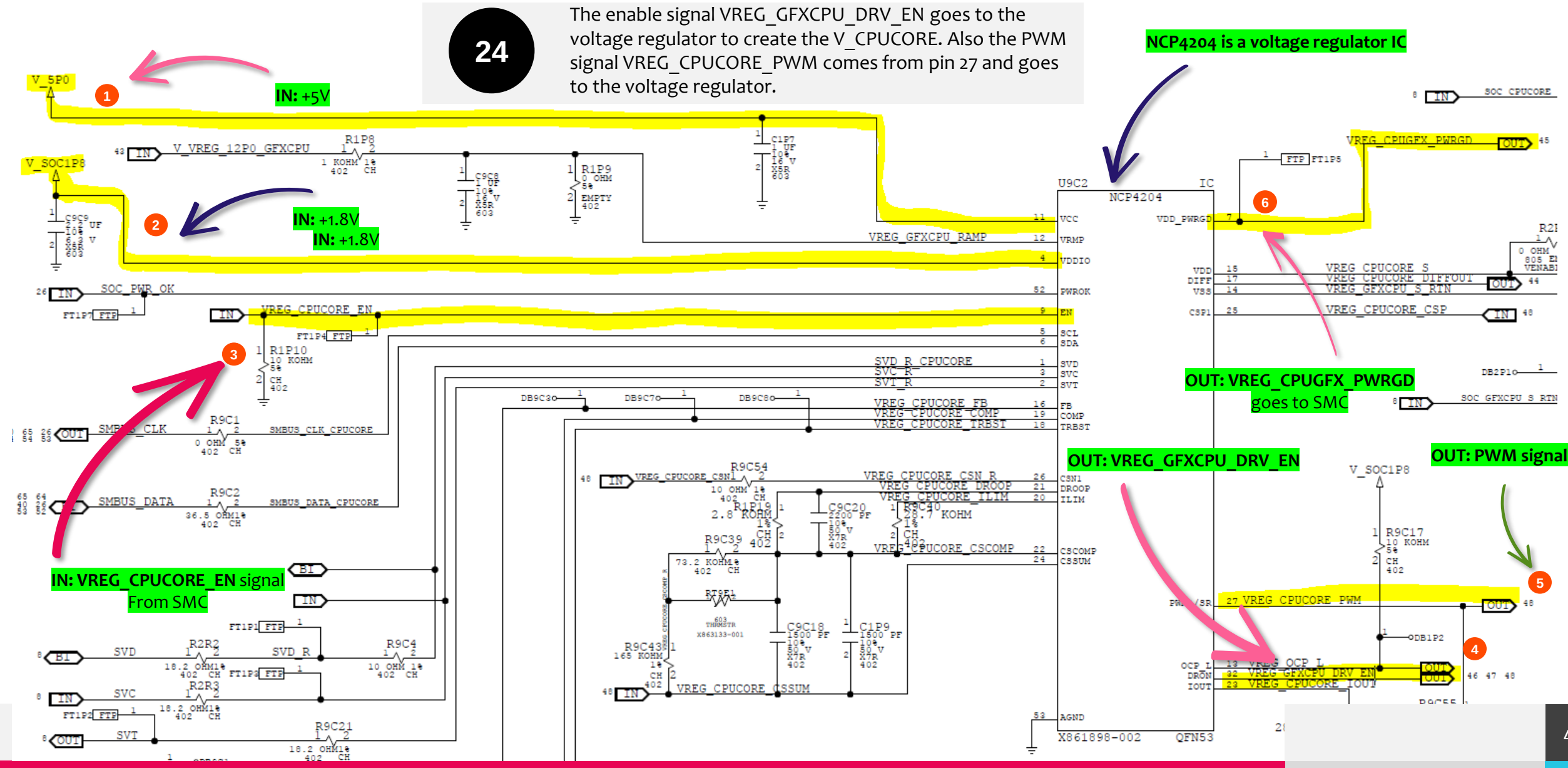
1 = BI: SVD – SoC <> NCP
2 = OUT: SVT - To SoC
3 = IN: SVC – From Soc
4 = IN: VDDIO – From V_SOC1P8
5 = OUT: SCL (clock)
6 = BI: SDA (I2C)
7 = OUT: VDD_PWRGD – To Southbridge
8 = OUT: VDDNB_PWRGD – To Southbridge
9 = IN: EN – From Southbridge
10 = IN: EN_NB – From Southbridge
11 = IN: VCC – From V_5P0
12 = IN: VRMP – From +12V
13 = OUT: OCP_L
14 = IN: VSS – From SoC
15 = IN: VDD – From SoC
16 = IN: FB – From pin 17
17 = OUT: DIFF - To Pin 16
18 = IN: TRBST
19 = IN: COMP
20 = IN: ILIM
21 = IN: DROOP
22 = IN: CSCOMP
23 = OUT: IOUT
24 = IN: CSSUM – total current sensing
25 = IN: CSP1 – From ouput V_CPUCORE
26 = IN: CSN1 – From ouput V_CPUCORE
27 = OUT: PWM1 – To CPUCORE voltage regulator IC
28 = OUT: PWM4NB – To GFXCORE voltage regulator IC phase 4

29 = OUT: PWM2NB – To GFXCORE voltage regulator IC phase 2
30 = OUT: PWM3NB – To GFXCORE voltage regulator IC phase 3
31 = OUT: PWM1NB – To GFXCORE voltage regulator IC phase 1
32 = OUT: DRON – To CPUCORE and GFXCORE enable signal
33 = IN: CSP1NB – From GFXCORE output phase 1
34 = IN: CSN1NB –From GFXCORE output phase 1
35 = IN: CSP3NB – From GFXCORE output phase 3
36 = IN: CSN3NB – From GFXCORE output phase 3
37 = IN: CSP2NB – From GFXCORE output phase 2
38 = IN: CSN2NB – From GFXCORE output phase 2
39 = IN: CSN4NB – From GFXCORE output phase 4
40 = IN: CSP4NB – From GFXCORE output phase 4
41 = IN: CSREFNB – From PIN 41
42 = IOUTNB
43 = IN: CSSUMNB - From GFXCORE output phase 1,2,3,4
44 = IN: CSCOMPNB
45 = IN: DROOPNB
46 = IN: ILMNB
47 = IN: COMPNB
48 = IN: FBNB
49 = IN: TRBSTNB
50 = OUT: DIFFNB – Goes to PIN 47, 48, 49
51 = IN: VDDNB – From SoC
52 = IN: PWROK – Comes from Southbridge

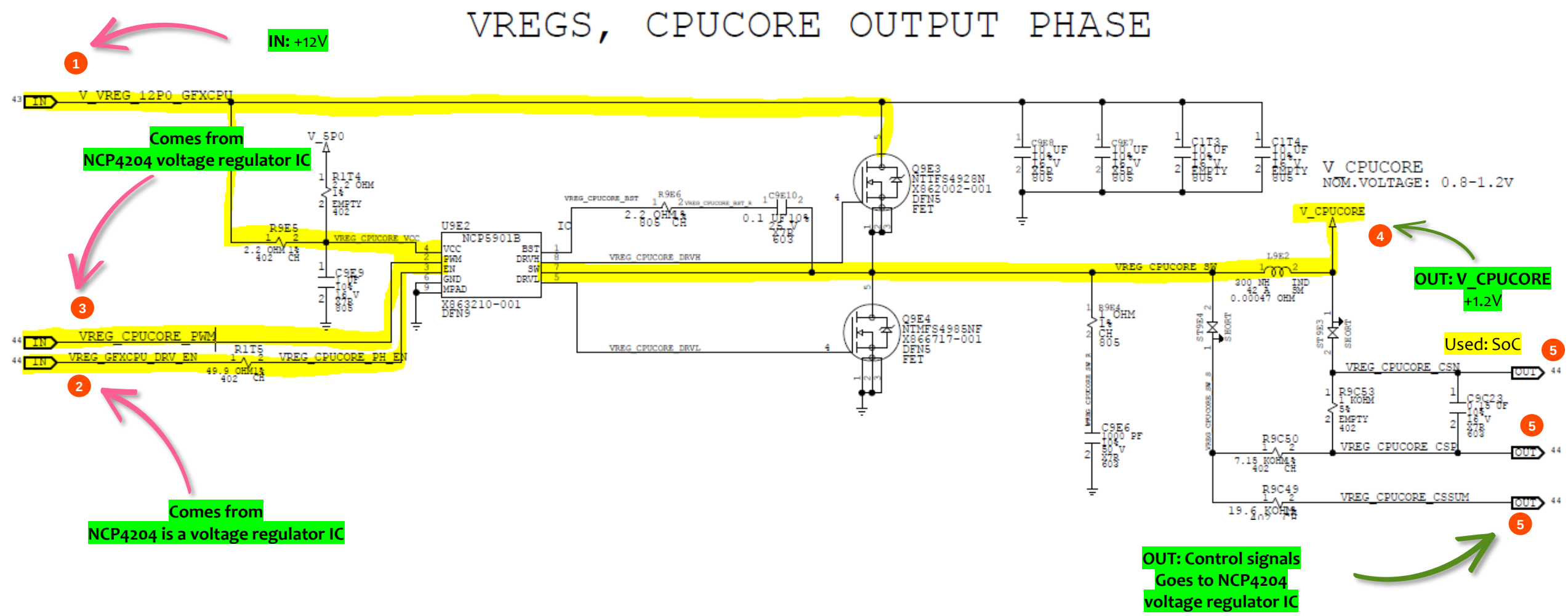
Step 24: Create PWM and EN signal for +1.2V regulator (V_CPUCORE)

24

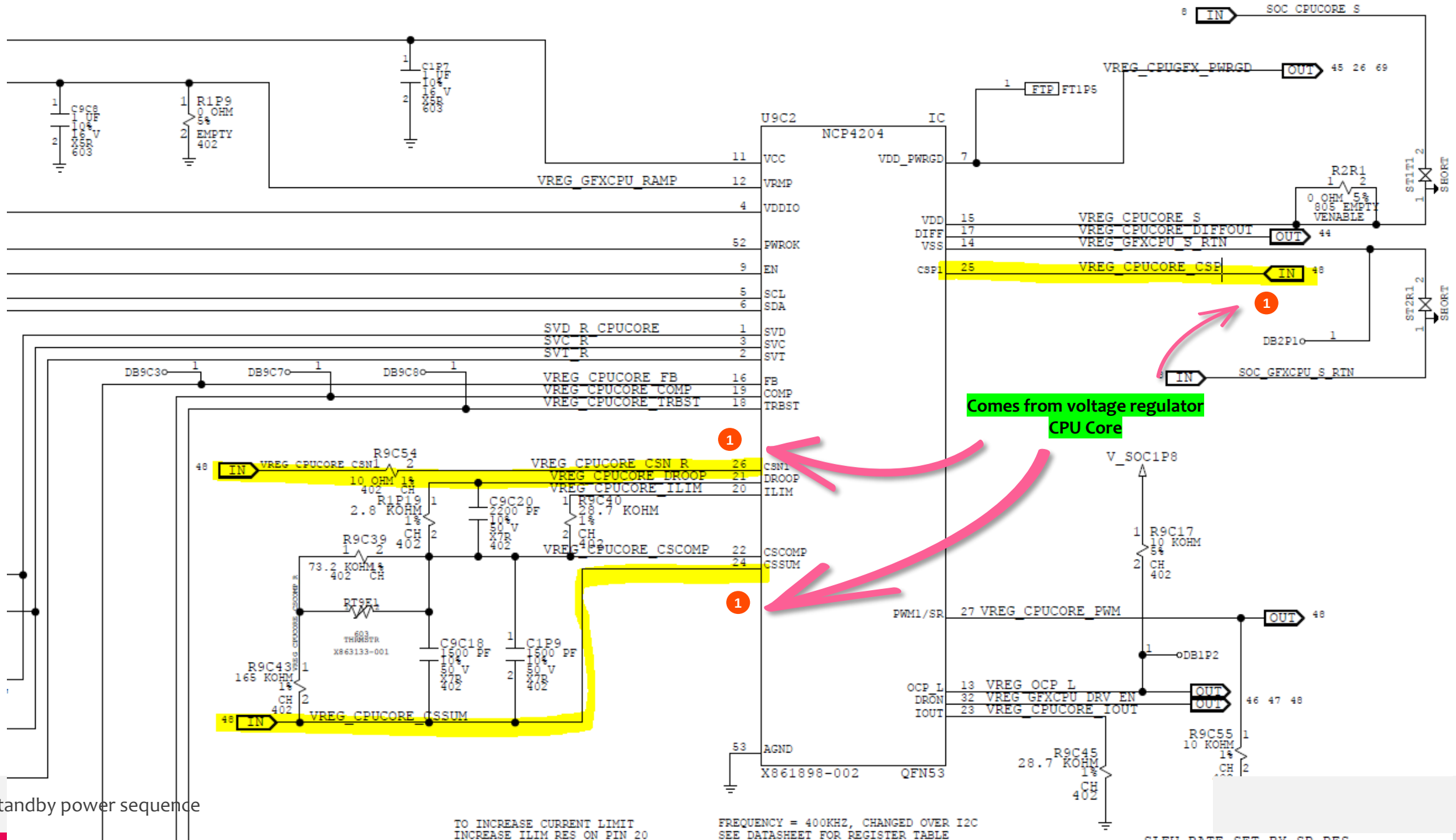
The enable signal VREG_GFXCPU_DRV_EN goes to the voltage regulator to create the V_CPUCORE. Also the PWM signal VREG_CPUCORE_PWM comes from pin 27 and goes to the voltage regulator.



Step 25: Voltage regulator will create power +1.2V (V_CPUCORE)

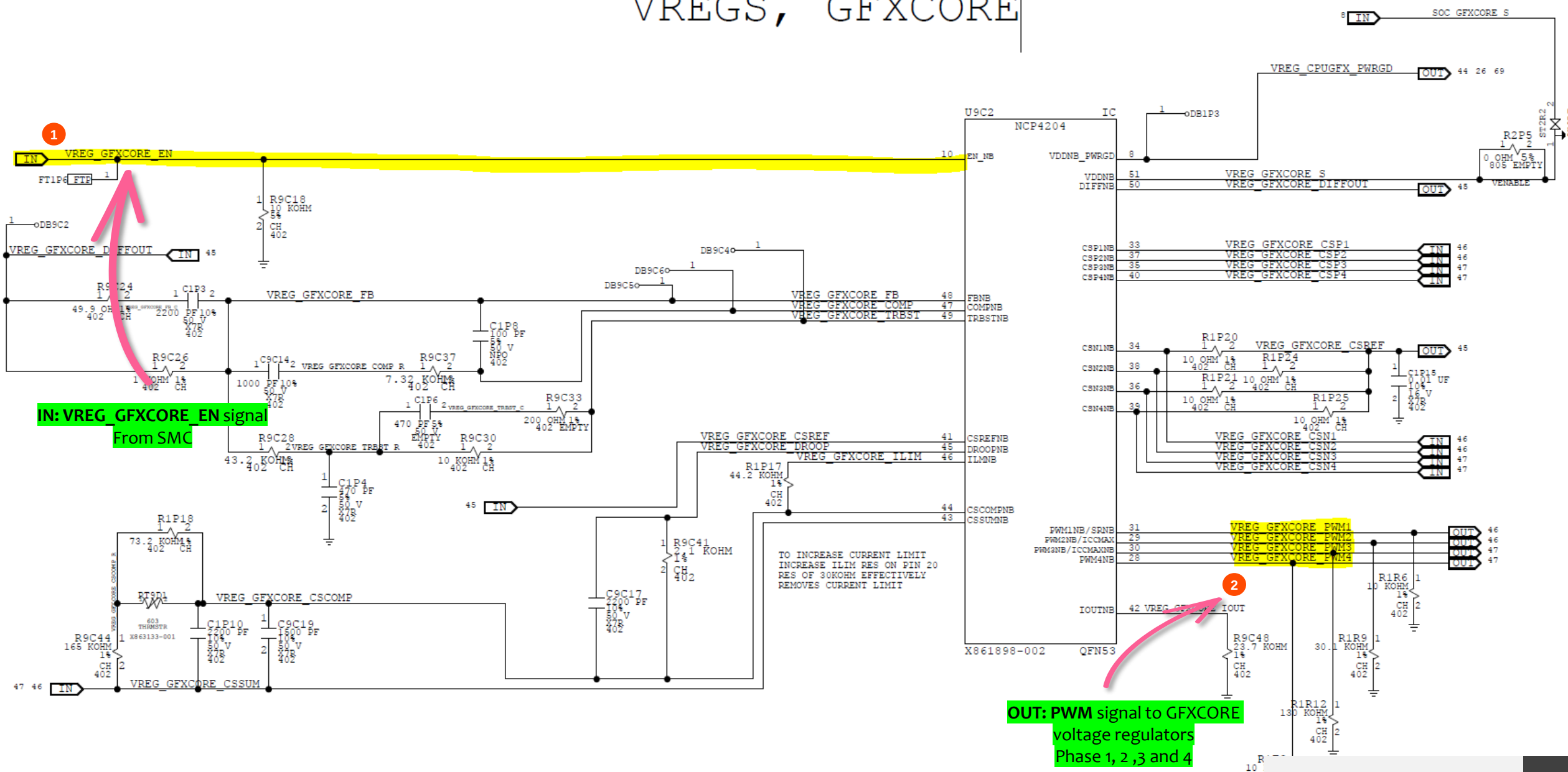


Step 26: V_CPUCORE Voltage regulator control signals

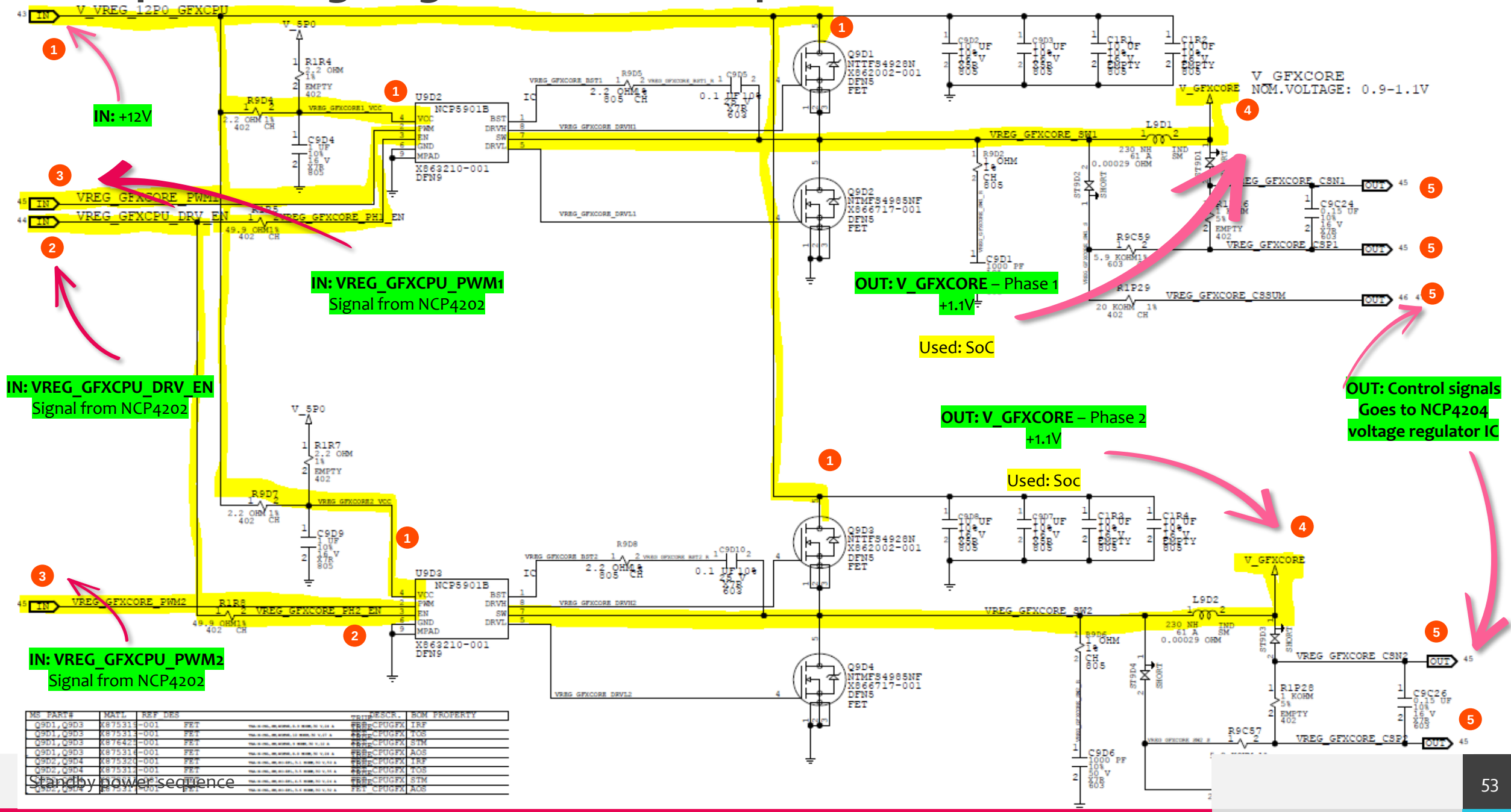


Step 27: Voltage regulator will create power +1.1V (V_GFXCORE) (1-2-3-4)

VREGS, GFXCORE

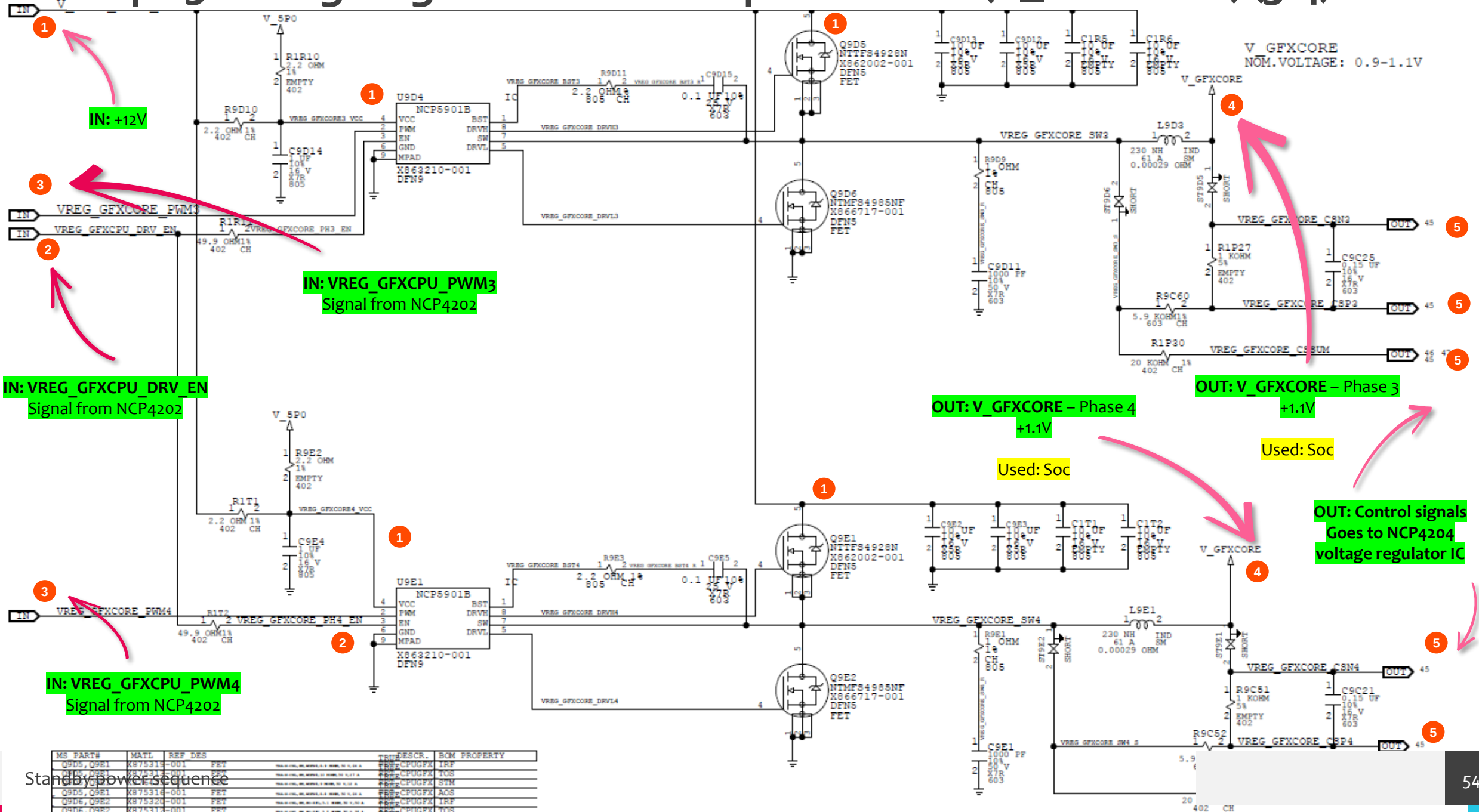


Step 28: Voltage regulator will create power +1.1V (V_GFXCORE) (1-2)



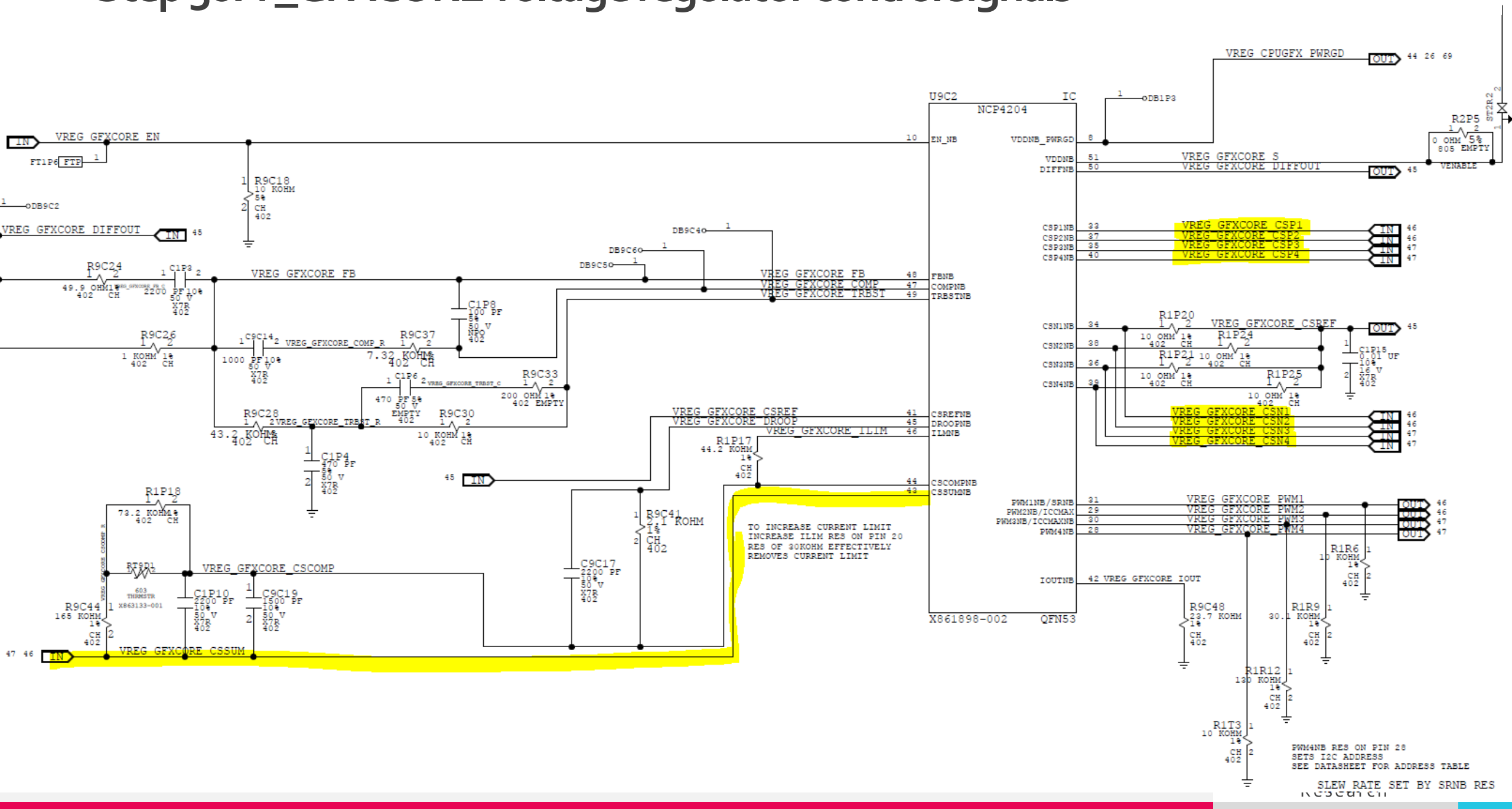
MS PART#	MATL	REF DES	DESCR.	BOM PROPERTY
Q9D1,Q9D3	X875319-001	FET		IRF
Q9D1,Q9D3	X875313-001	FET		TOS
Q9D1,Q9D3	X876425-001	FET		STM
Q9D1,Q9D3	X875316-001	FET		AOS
Q9D2,Q9D4	X875320-001	FET		IRF
Q9D2,Q9D4	X875312-001	FET		TOS
Q9D2,Q9D4	X875317-001	FET		STM
Q9D2,Q9D4	X875314-001	FET		AOS

Step 29: Voltage regulator will create power +1.1V (V_GFXCORE) (3-4)

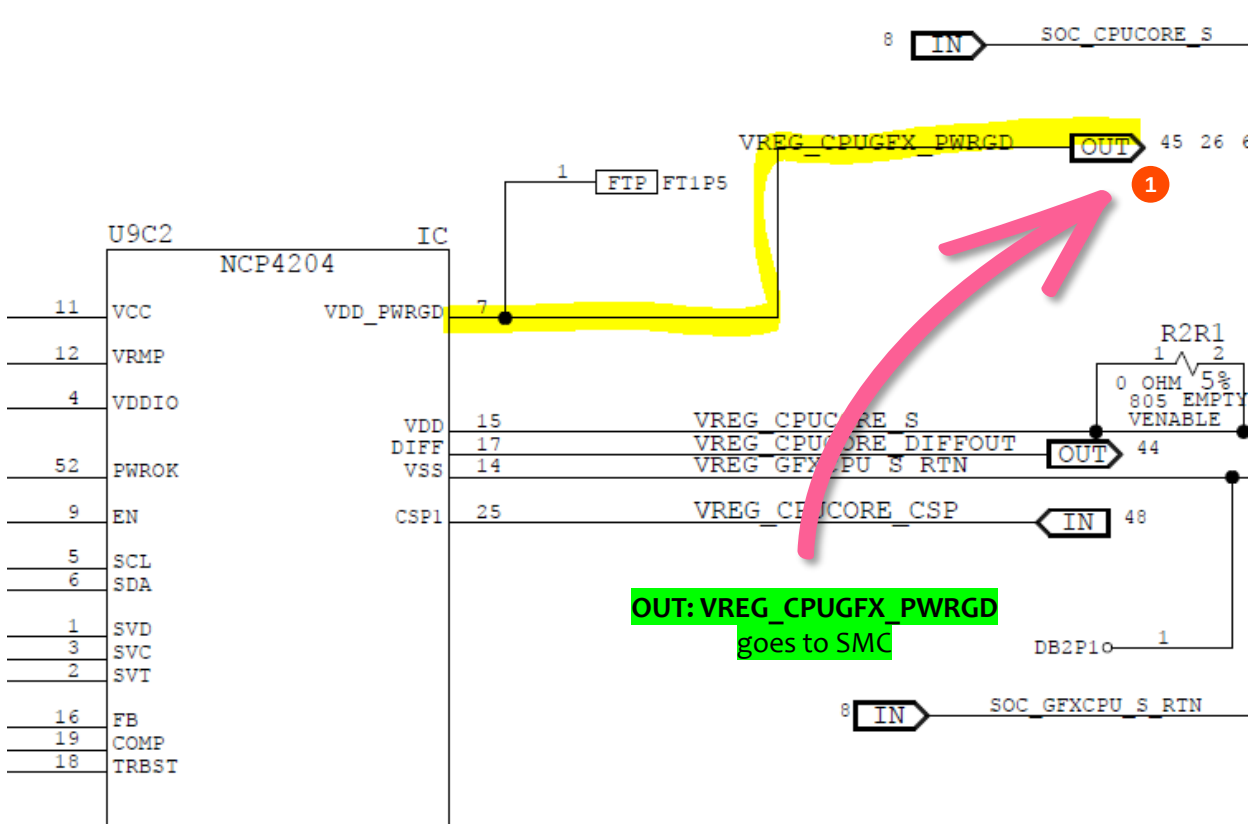


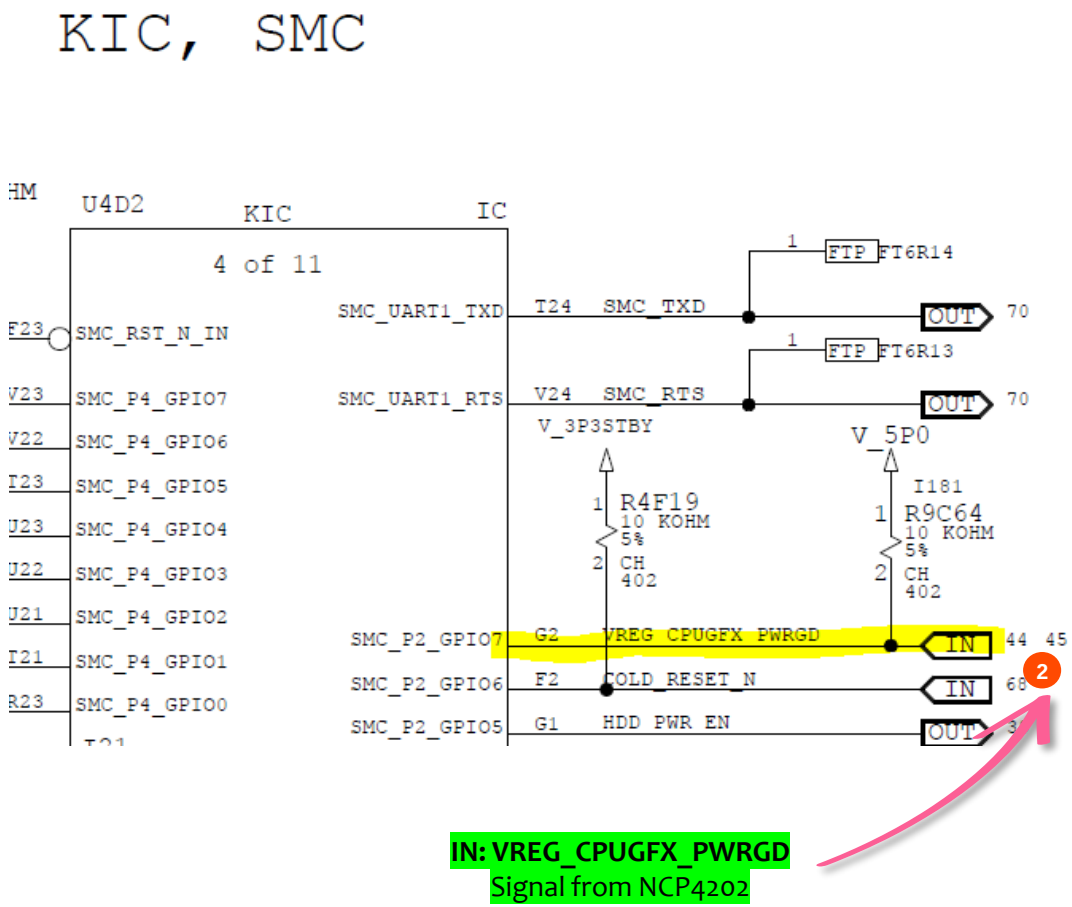
MS_PART#	MATL	REF_DES	DESCR.	BOM_PROPERTY
Q9D5,Q9E1	X875319-001	FET	CPUGFX	IRF
Q9D6,Q9E2	X875320-001	FET	CPUGFX	TOS
Q9D5,Q9E1	X875316-001	FET	CPUGFX	STM
Q9D6,Q9E2	X875320-001	FET	CPUGFX	AOS
Q9D5,Q9E1	X875316-001	FET	CPUGFX	IRF
Q9D6,Q9E2	X875320-001	FET	CPUGFX	TOS

Step 30: V_GFXCORE Voltage regulator control signals

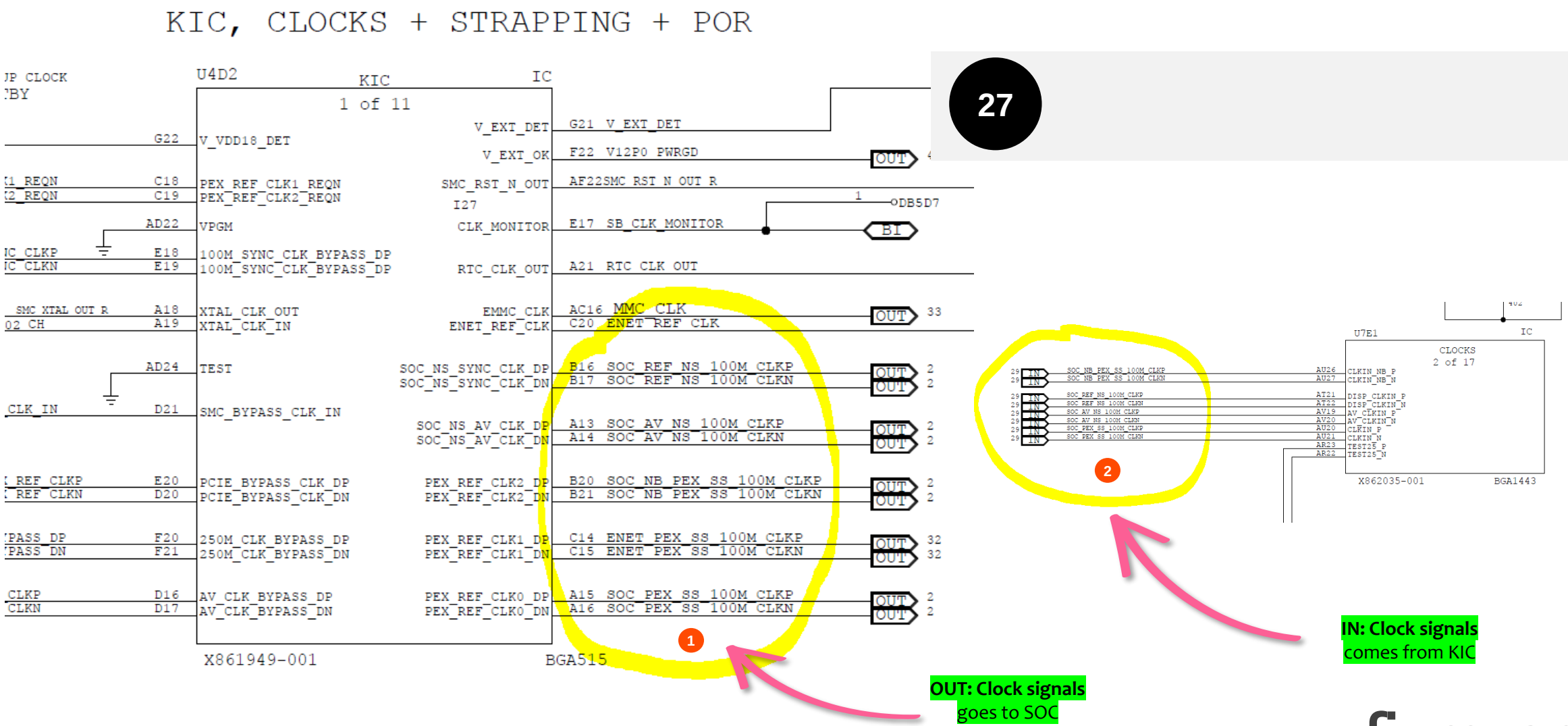


Step 31: Send power good signal to SMC – VREG_CPUGFX_PWRGD

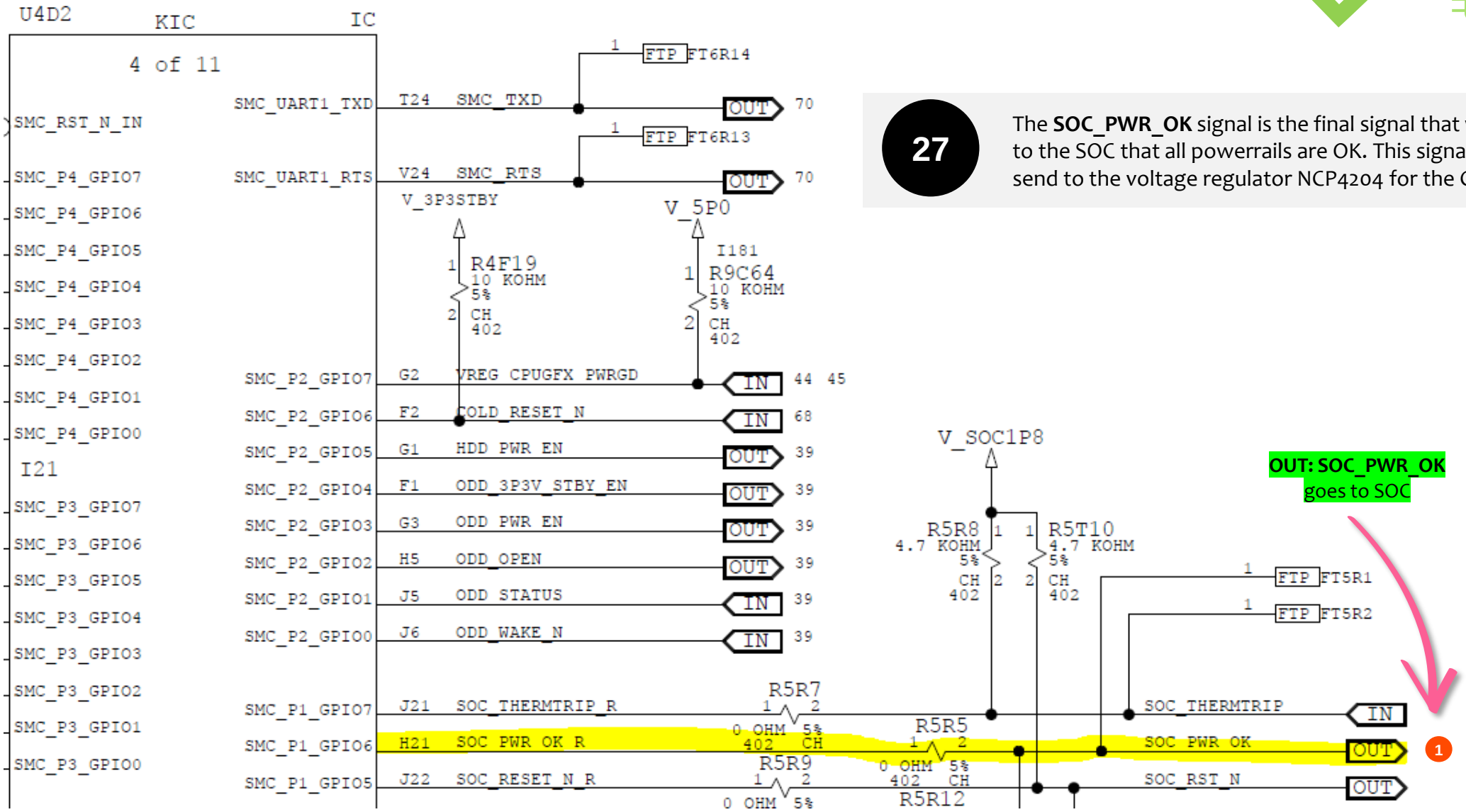




Step 32: KIC will generate 100Mhz clock signal to SOC



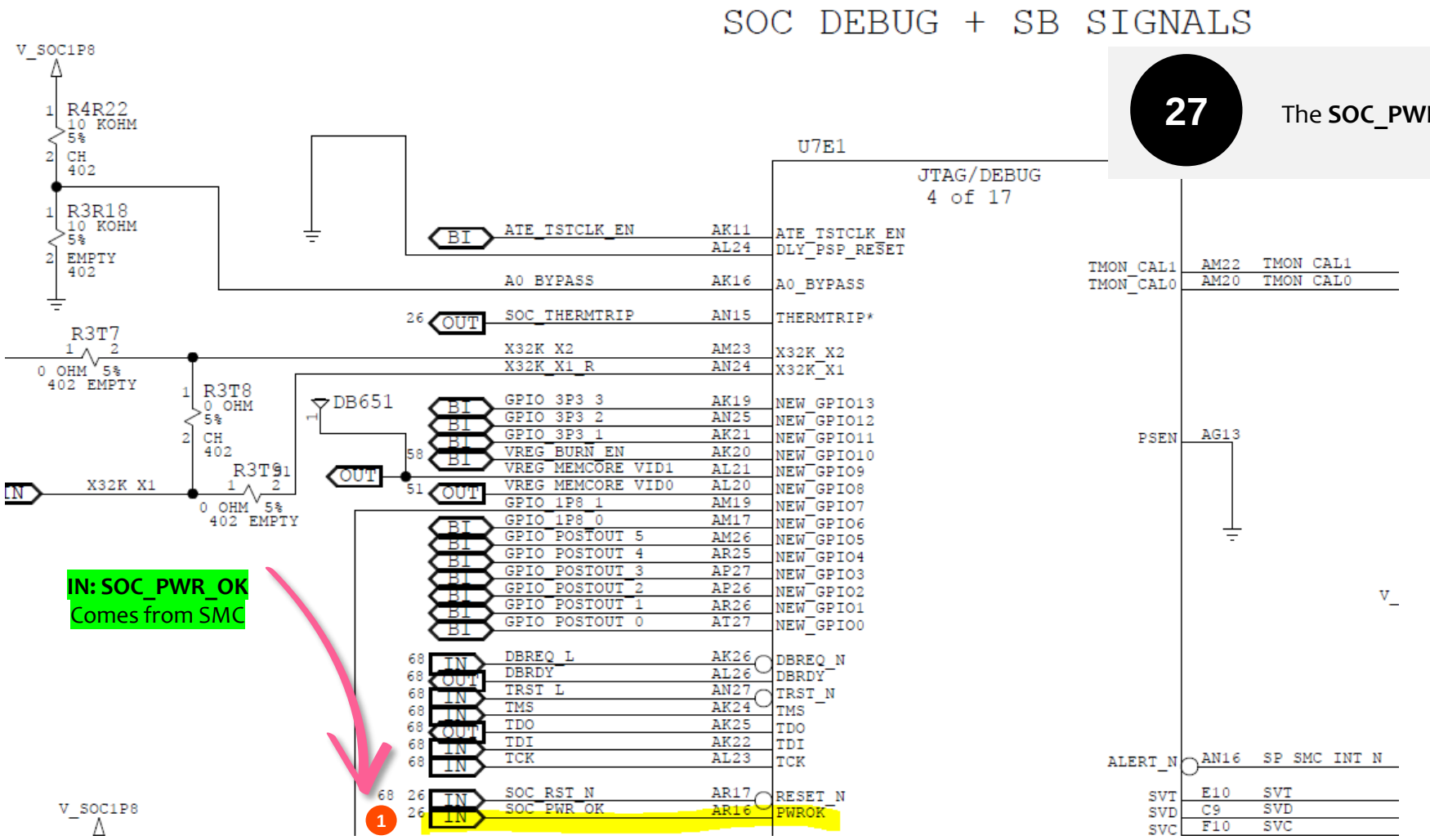
FINAL STEP: SMC will generate final power good signal:



27

The **SOC_PWR_OK** signal is the final signal that will be send to the SOC that all powerrails are OK. This signal is also send to the voltage regulator NCP4204 for the CPUCORE

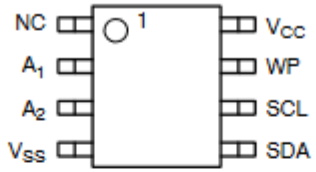
FINAL STEP : SOC will receive the final power good signal:



27 The **SOC_PWR_OK** signal received by the SOC.

BIOS IC – CAT24M01

PIN CONFIGURATION

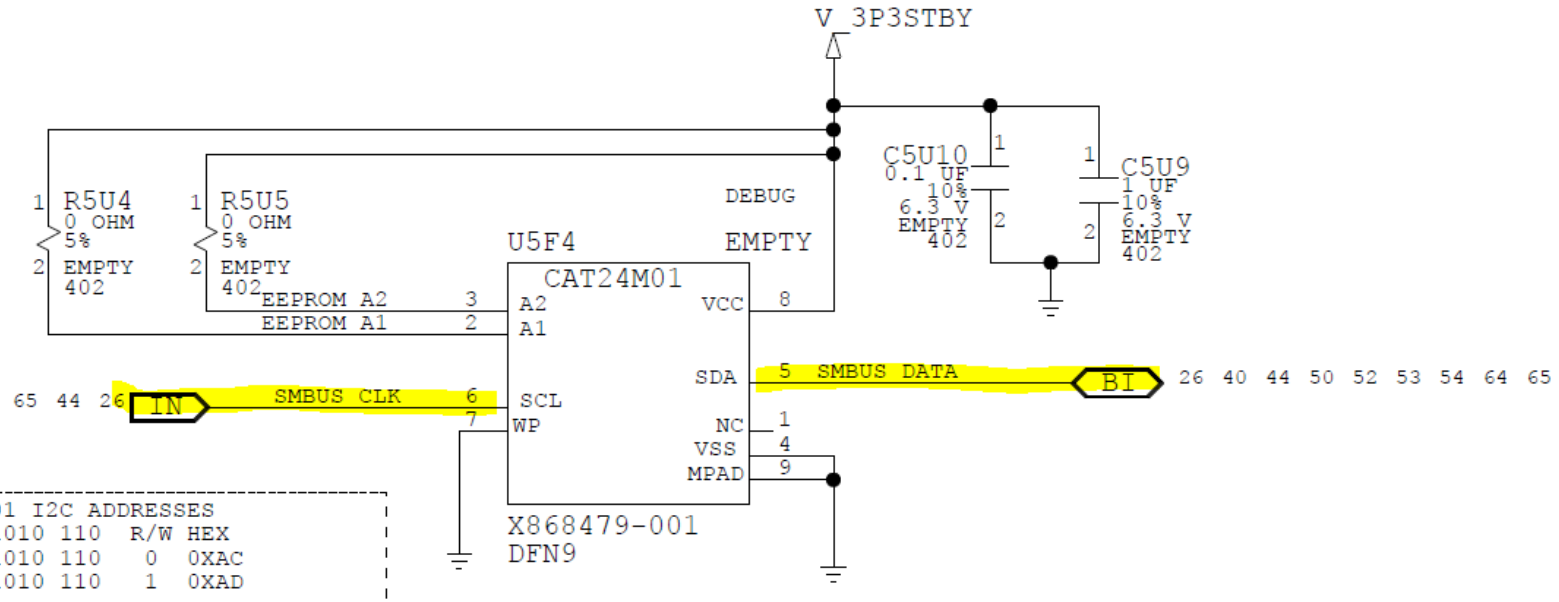


SOIC (W, X), TSSOP (Y),
UDFN (HU5)

For the location of Pin 1, please consult the corresponding package drawing.

PIN FUNCTION

Pin Name	Function
A_1, A_2	Device Address
SDA	Serial Data
SCL	Serial Clock
WP	Write Protect
V_{CC}	Power Supply
V_{SS}	Ground

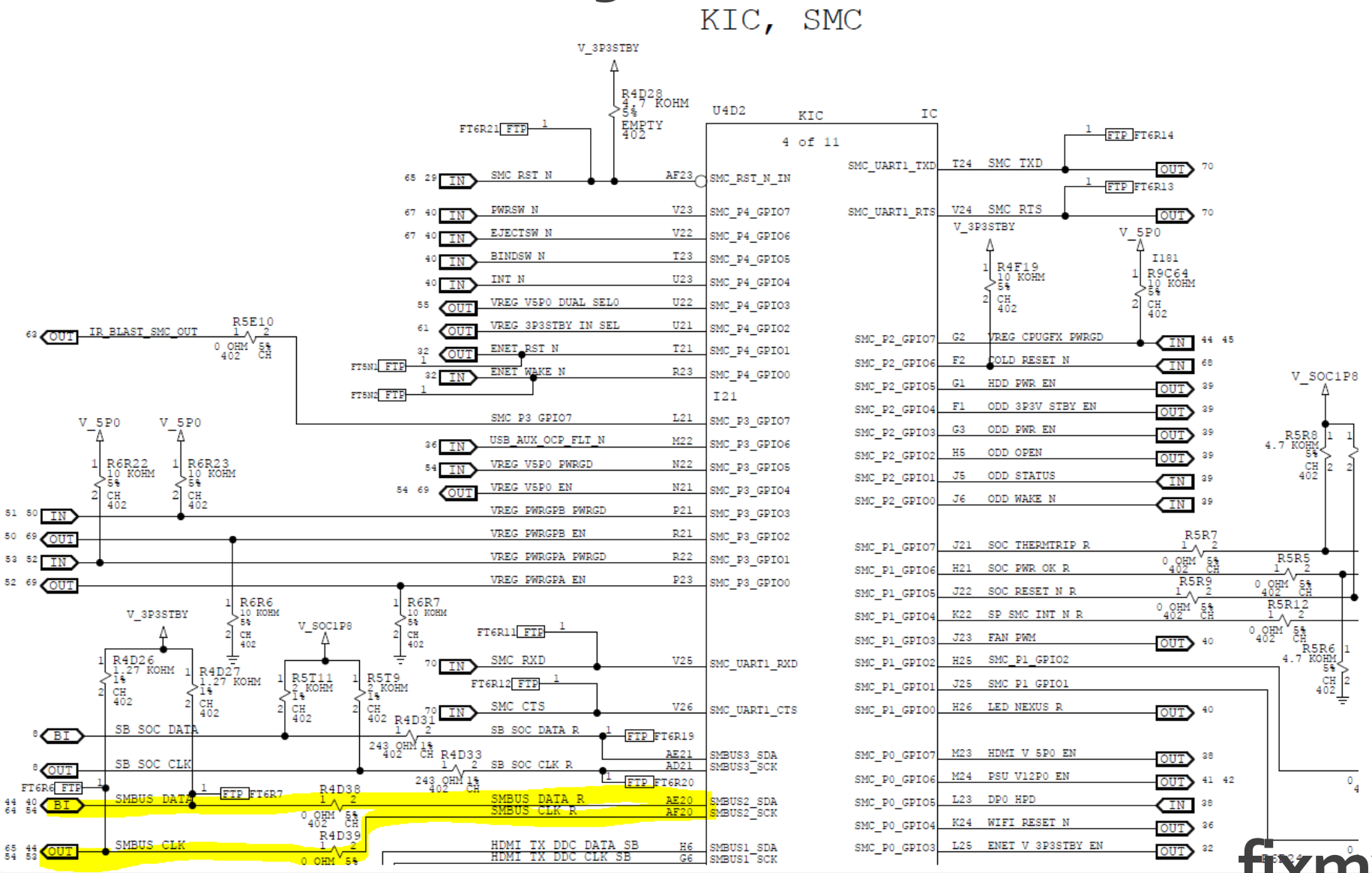



```
CAT24M01 I2C ADDRESSES
       1010 110  R/W HEX
WRITE  1010 110    0  0XAC
READ   1010 110    1  0XAD

       1010 111  R/W HEX
WRITE  1010 111    0  0XAE
READ   1010 111    1  0XAF
```

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BIOS IC – Clock and Data – Southbridge IC



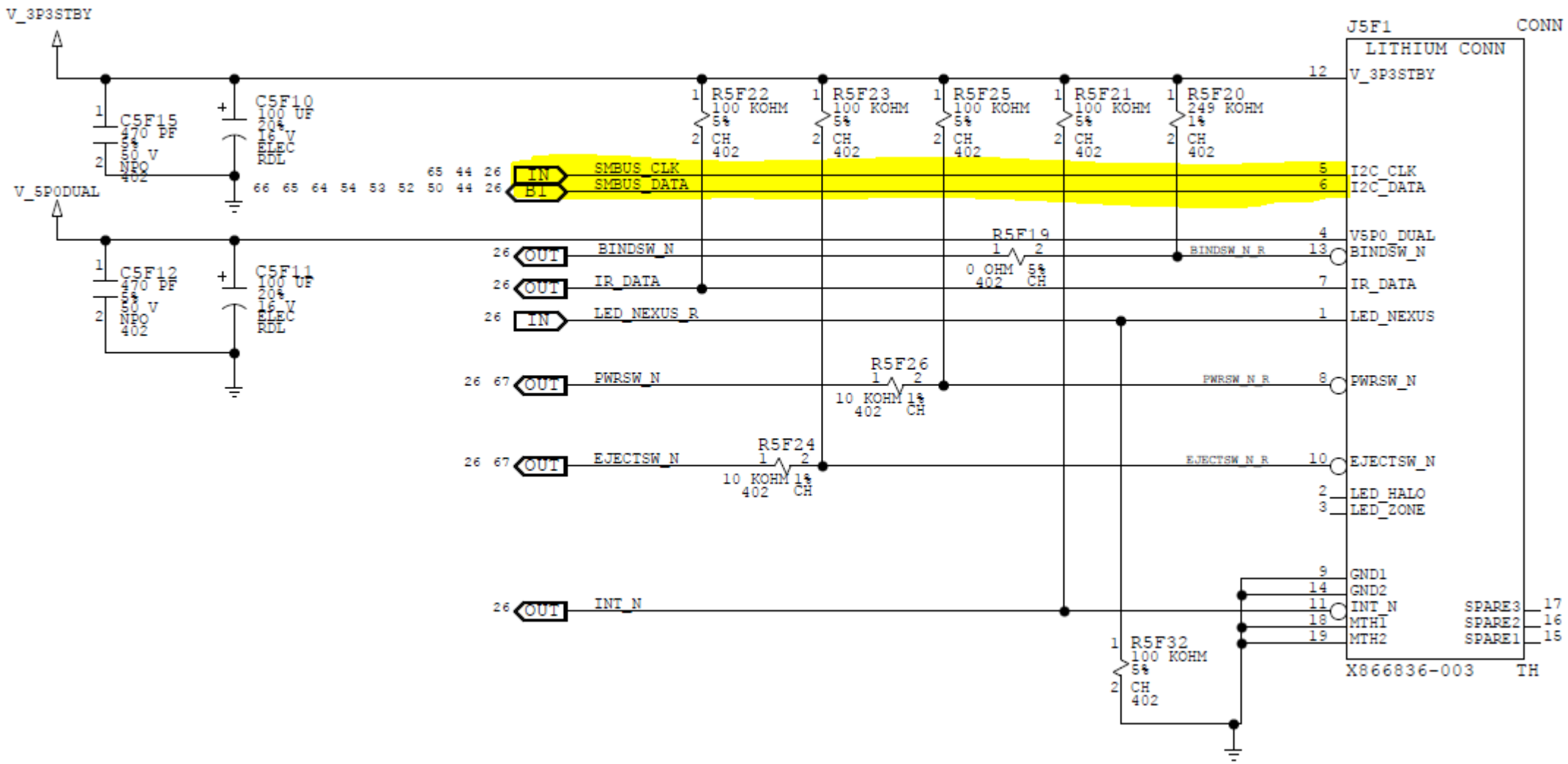
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PMBus voltage regulator command

Table 3. Supported PMBus Commands

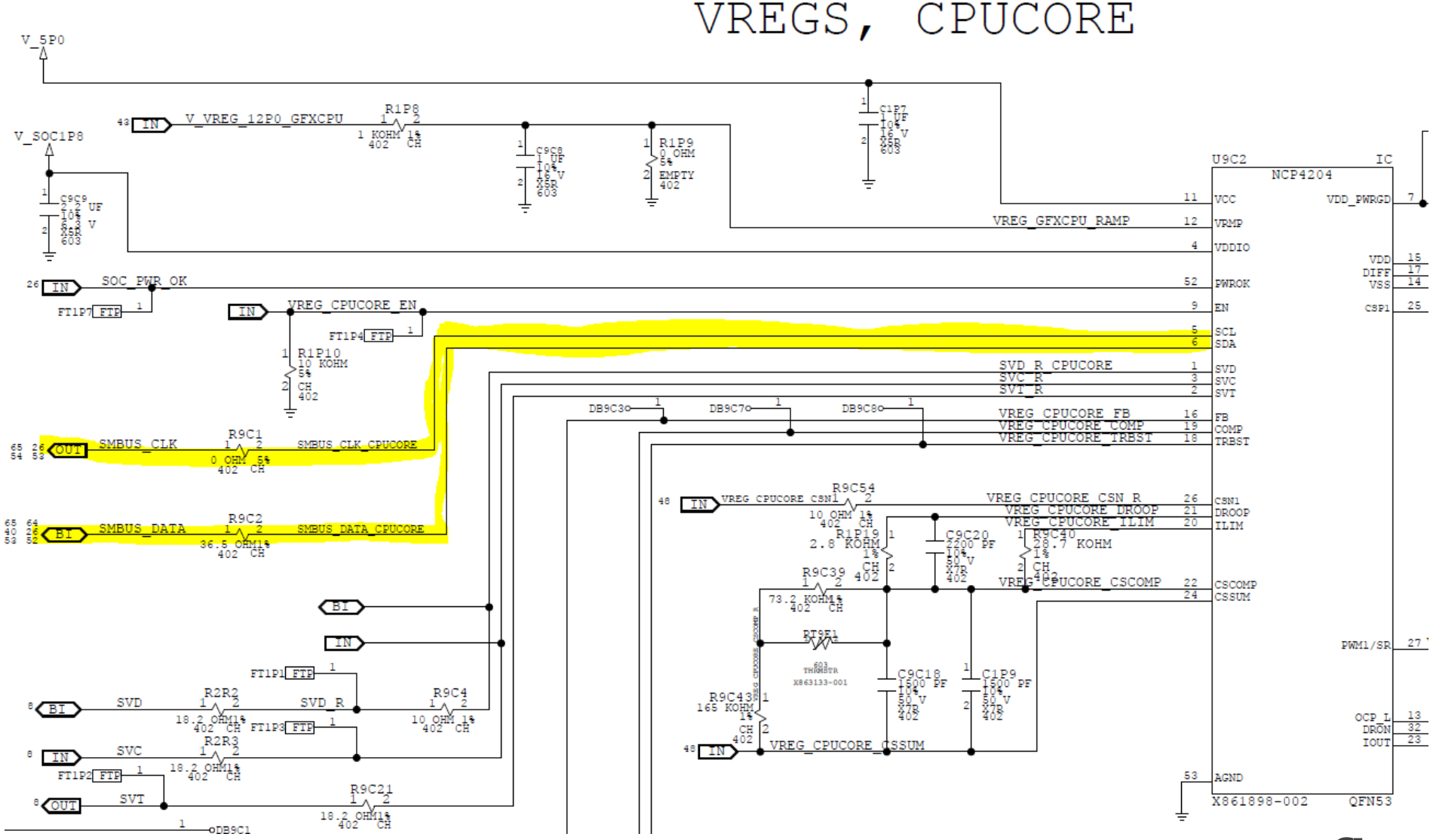
COMMAND	NOTES
OPERATION	Turn on or turn off switching converter only
ON_OFF_CONFIG	ON/OFF configuration
CLEAR_FAULTS	Clear all latched status flags
WRITE_PROTECT	Control writing to the PMBus device
STORE_DEFAULT_ALL	Store contents of user-accessible registers to non-volatile memory cells
RESTORE_DEFAULT_ALL	Copy contents of non-volatile memory cells to user-accessible registers
STATUS_WORD	PMBus read-only status and flag bits
CUSTOM_REG	MFR_SPECIFIC_00 (Custom Register 0): Custom register
DELAY_CONTROL	MFR_SPECIFIC_01 (Custom Register 1): Power on and power good delay times
MODE_SOFT_START_CONFIG	MFR_SPECIFIC_02 (Custom Register 2): Mode and soft-start time
FREQUENCY_CONFIG	MFR_SPECIFIC_03 (Custom Register 3): Switching frequency control
VOUT_ADJUSTMENT	MFR_SPECIFIC_04 (Custom Register 4): Output voltage adjustment control
VOUT_MARGIN	MFR_SPECIFIC_05 (Custom Register 5): Output voltage margin levels
UVLO_THRESHOLD	MFR_SPECIFIC_06 (Custom Register 6): Turn-on input voltage UVLO threshold

BIOS IC – Clock and Data – Frontpanel



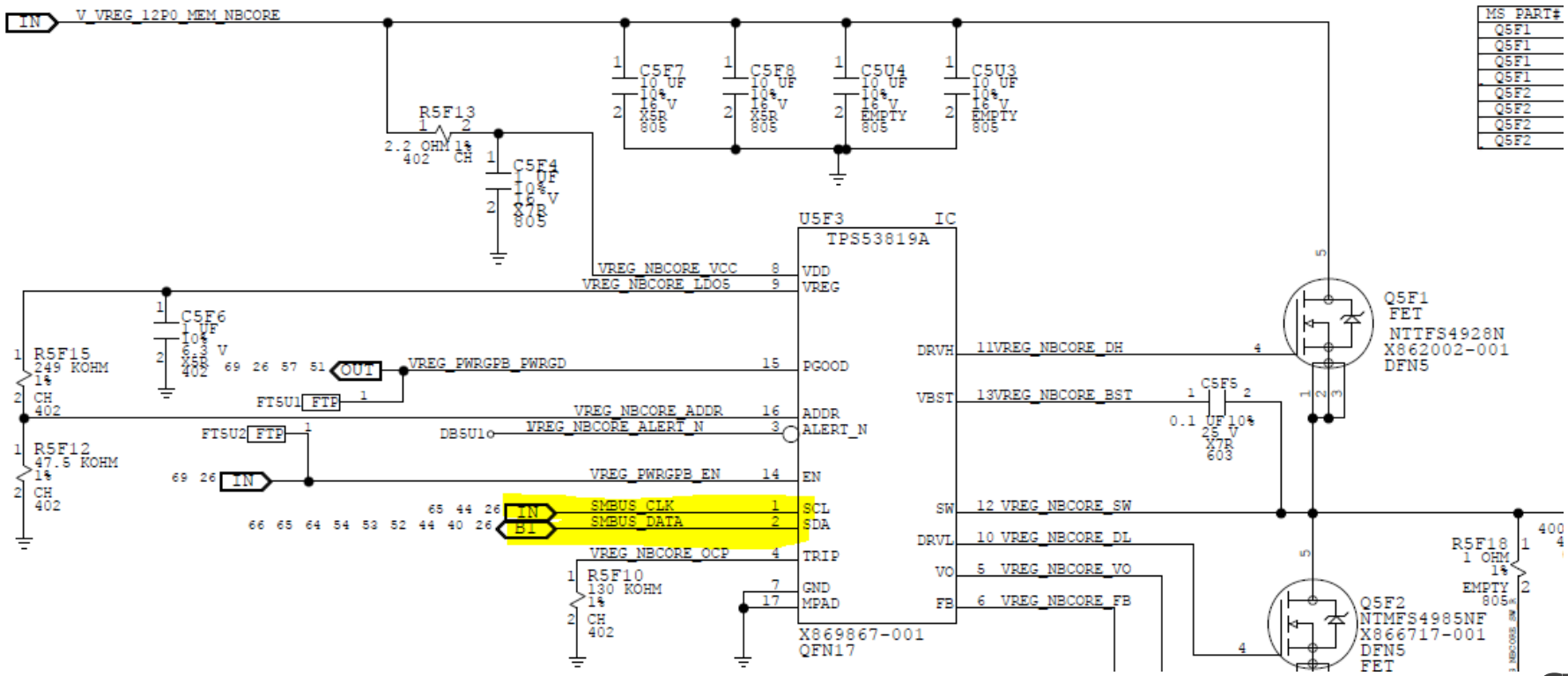
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BIOS IC – Clock and Data – NCP4202 – GFXCORE / CPUCORE Regulator



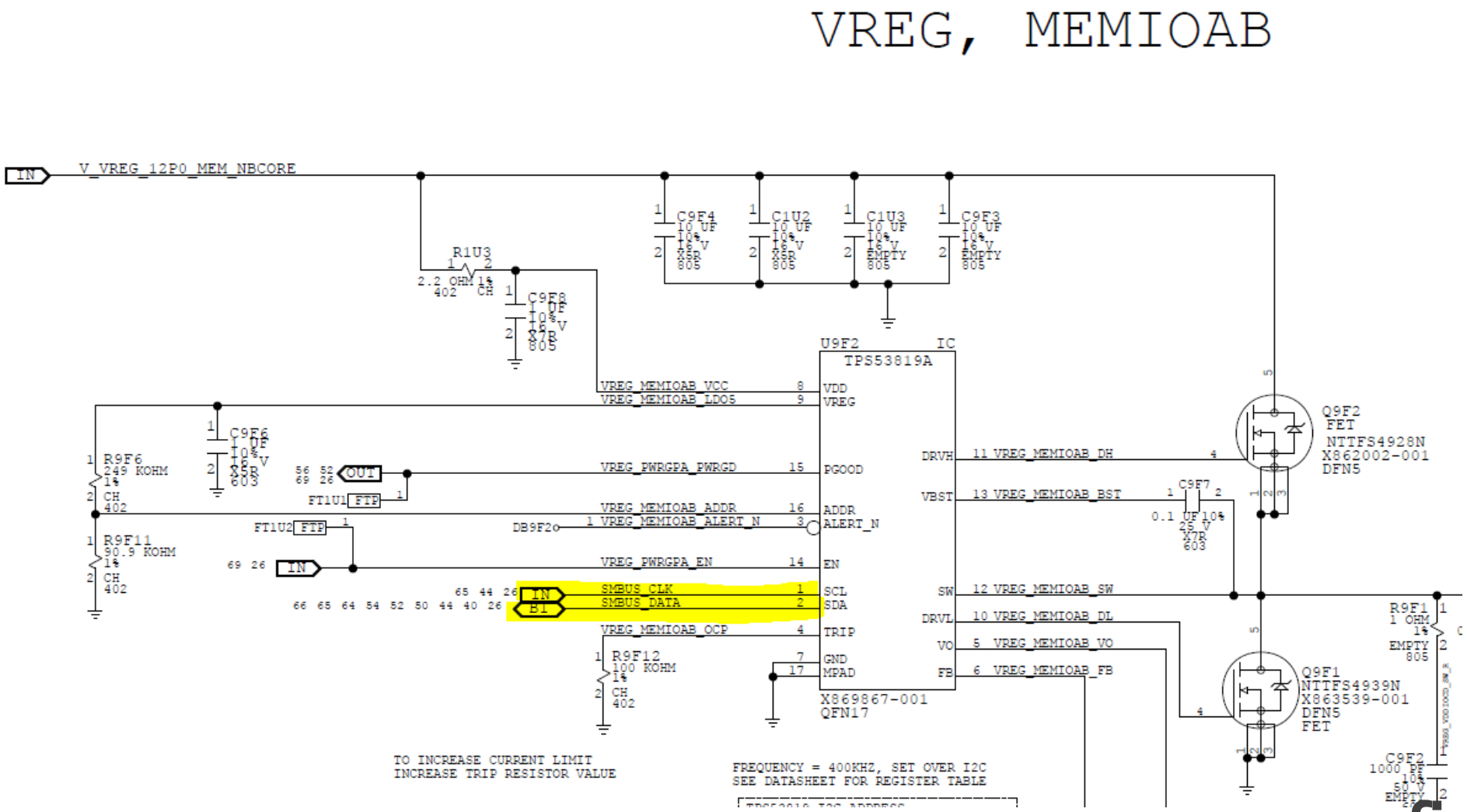
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BIOS IC – Clock and Data – NBCORE Regulator



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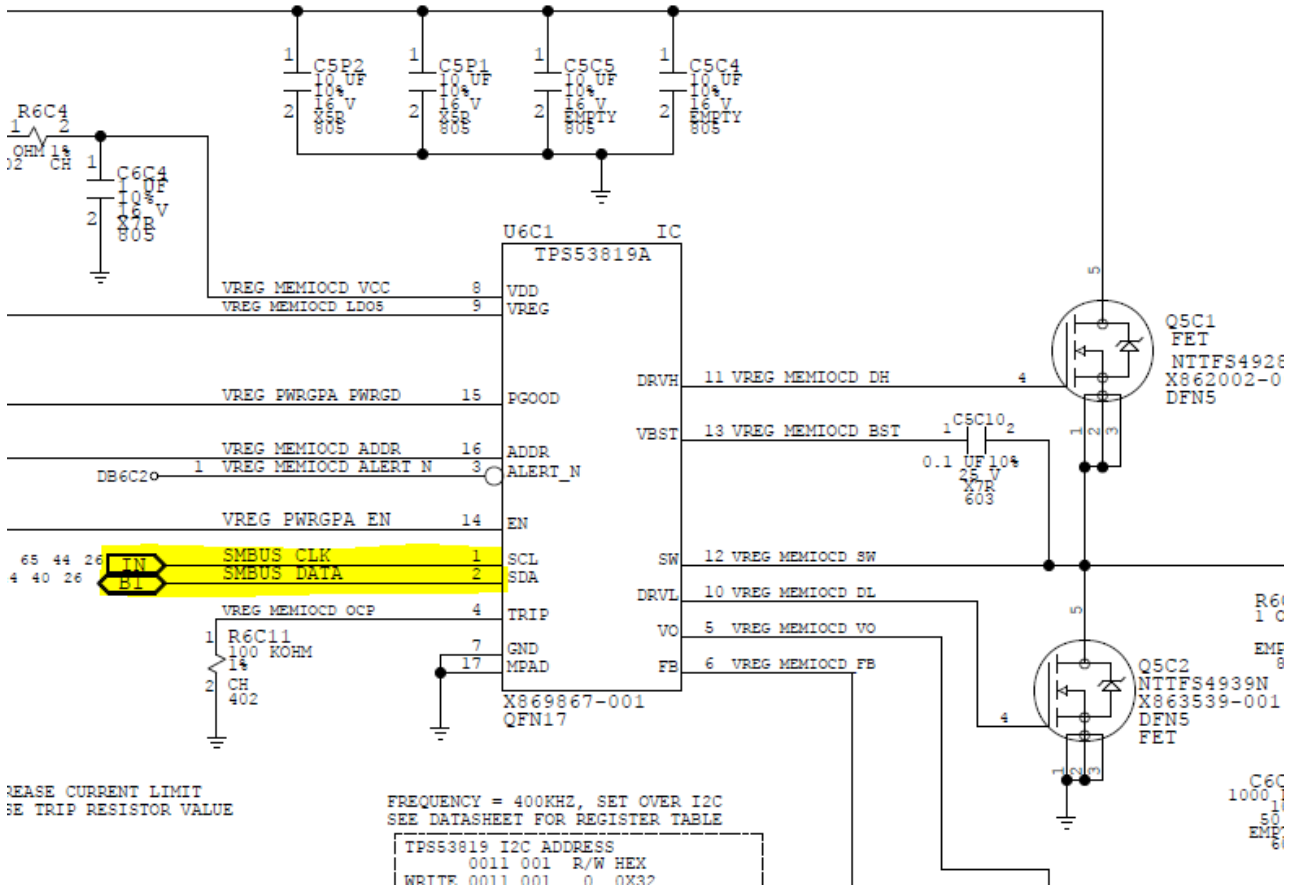
BIOS IC – Clock and Data – MEMIOAB Regulator



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BIOS IC – Clock and Data – MEMIOCD Regulator

VREG, MEMIOCD



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